Hyperspectral Unmixing via Noise-Free Model

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***Abstract*— Blind hyperspectral unmixing (BHSU) is ill-posedness. It aims to obtain accurate and robust endmember signatures and the corresponding abundances simultaneously. Nonnegative matrix factorization (NMF)-based sparsity-regularized algorithms have been widely employed for the BHSU. However, the existing unmixing approaches are sensitive to the multifarious intrinsic interferences and noises, which are caused because of the utilization of the inappropriate loss function to measure the quality of the hyperspectral data (HD) reconstruction and regularization. In this article, we propose a noise-free graph regularized model (NFGRM) by applying the dual graph regularized robust nonnegative matrix tri-factorization (NMTF), which leads to a novel reliable reconstruction of the HD. In the NFGRM, all the challenging interferences are addressed as noises. Consequently, a more faithful approximation is expected to recover from the highly noisy mixed data set and achieve robust regularization by controlling the heteroscedastic noises and the ill-posedness of the BHSU problem simultaneously. Experimental results on synthetic and several benchmark HD sets demonstrate the effectiveness and robustness of the proposed model and algorithm.**

***Index Terms*— Blind hyperspectral unmixing (BHSU), graph dual regularization, noise-free, nonnegative matrix tri-factorization (NMTF).**

NOMENCLATURE

Throughout this article, we use $\mathrm{Tr}(\cdot)$ to represent the trace of a matrix, to denote the element-wise product and element-wise division via $\odot$ and $\oslash$. $()^{\cdot-(1/2)}$ denotes elementwise power of $-(1/2)$. $(\cdot)^{\mathrm{T}}$ is the transposition of a vector or a matrix, $(\cdot)^{-1}$ calculates the inverse of a matrix, $\|\cdot\|_{(1/2)}$ denotes $L_{(1/2)}$ norm. diag($\mathbf{Z}$) denotes a diagonal matrix of $\mathbf{Z}$.

BHSU	Blind hyperspectral unmixing.
NMF	Nonnegative matrix factorization.
HD	Hyperspectral data.
NFGRM	Noise-free graph regularized model.
NMTF	Nonnegative matrix tri-factorization.
LMM	Linear mixing model.
NLMM	Nonlinear mixing model.
MV	Minimum-volume.
EV	Endmember variability.
SED	Square of Euclidean distance.
SCLSU	Scaled constrained least squares unmixing.
SAD	Spectral angle distance.
RMSE	Root mean square error.
GPSF	Gaussian point spread function.
SNR	Signal-to-noise ratio.
$\mathbf{Y} \in \Re^{M\times P}$	Matrix of a HD, where M and P denote the number of spectral bands and pixels in the HD.
$\mathbf{S} \in \Re^{M\times D}$	Endmember matrix that contains D endmember signatures.
$\mathbf{U}\in\Re^{D\times P}$	Abundance matrix.
$\mathbf{E} \in \Re^{M\times P}$	Reconstruction error matrix.
$\mathbf{V} \in \Re^{D\times D}$	Data-dependent matrix to mitigate the EV.
$\mathbf{L_S}$, $\mathbf{L_U}$	Laplacian matrices on the spectral graph and pixel graph, respectively, and $\mathbf{L}_X = \mathbf{L}_U$.
$\mathbf{W_S}$, $\mathbf{W_U}$	Manifold weight matrices on the endmember graph and pixel graph, and $\mathbf{W}_X = \mathbf{W}_U$.
$\mathbf{X} \in \Re^{D\times P}$	Pixel cluster assignment matrix.
$\mathbf{Z} \in \Re^{D\times D}$	Dummy variable, and $\mathbf{Z} = \mathbf{V}^{-1}$.
$\mathbf{E}_1$, $\mathbf{E}_2$	Row and column standard deviations of the matrix $\mathbf{E}$.
Υ, Φ, Ψ, Θ	Corresponding Lagrange multipliers of the parameters $\mathbf{S}$, $\mathbf{Z}$, $\mathbf{X}$ and $\mathbf{E}$.
σ_d, λ_d	Standard deviations of the heteroscedastic noise and eigenvalues of the endmember matrix.
η	η is a constant to adjust the proportion of the construction error in the HD.
β_2, β_3	Tradeoff parameters for the graph regularizer.
β_1, β_4	Tradeoff parameters for the sparse constraints.
β_5	Constant to control the initial value for $\mathbf{Z}$.
$\mathbf{B} \in \Re^{M\times M}$	Blurring matrix for simulating the nonlinearity.

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I. INTRODUCTION

DUE to the low spatial resolution of the HD, many pixels are acquired by the hyperspectral imaging sensor with several spectral signatures, that is, causing many "mixed pixels." These mixed pixels are considered to be the combination of unique spectra, called endmember signatures. The contribution proportion of an endmember signature to the spectrum of a mixed pixel is called abundance [1]. If the endmembers and abundance are unknown beforehand, the task of hyperspectral unmixing becomes BHSU, which is an important technology

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for simultaneously decomposing the mixed pixels into a series of spectra of pure substances and the corresponding fractional abundances [2].

Over the last few decades, an extensive number of different constrained BHSU approaches have been presented [3]–[7], and most of them are based on two broadly used mixing models, namely LMM and NLMM [3], [8]. The physical interpretation behind the LMM/NLMM is that a pixel is supposed to be a linear/nonlinear combination of endmember signatures weighted by their fractional abundances. Among the above-mentioned BHSU approaches, the endmember signatures and their fractional abundances are extracted and quantified with little prior knowledge. These BHSU methods belong to either geometrical-based methods or statistical methods [9]. The geometrical-based methods model the endmembers as the vertices of a simplex [10] and enclose the HD with the smallest volume, such as Craig-criterion-based MV enclosing simplex method [11], MV simplex analysis [12] and MV constrained NMF [13]. Although geometrical-based algorithms are usually more efficient than the statistical methods, the assumed simplex is sensitive to the artifacts and the outliers[1] [15].

NMF is one of the practical and powerful methods for the BHSU [9], which decomposes the HD into two low-rank nonnegative matrices. Since the decomposition of NMF is a nonconvex problem, the decomposition is unstable and very susceptible to the initialization. Some additional constraints are indispensable in the NMF-based methods, such as, sparsity-constrained deep NMF [16], smooth and sparse regularization for NMF [17], sparsity-constrained deep NMF with total variation [18], and blind sparse nonlinear unmixing using a l_q penalty [6]. Despite NMF-based unmixing methods have improved the performance of the BHSU, they still suffer from the following issues. First, most of these methods employ the squared loss function, which is a well-known criterion to measure the reconstruction error. While a few noisy elements with large errors may easily deteriorate the reconstruction [19], that is, the squared loss function is unreliable with respect to abrupt noises and outliers. Second, these NMF-based methods generally consider pixel side clustering and neglect the cluster structure on spectral side [20]. Practically, clustering the endmembers and clustering abundances are closely related in the BHSU. However, unlike NMF, NMTF has been proven to be scalable. It is an effective method for sparse data [21] by improving clustering along both dimensions and achieving an implicit adaptive dimensionality reduction for each stage. Third, the sparsity-constrained NMF-based methods have been based on l_0 sparsity regularizer, which are NP-hard problem. In addition, they do not account for the heterogeneous structure of the HD (the varied surface, illumination source, and atmospheric conditions can bring about many interferences to the HD [22], such as, EV may occur in every pixel of the HD [23], [24], and the nonlinearity usually causes the heterogeneous structure of the HD), and depend on the unknown entries to be estimated. Fourth, when the HD is contaminated by noises and outliers, the constructed sparse constraint may also be corrupted and fail to acquire the desired structure information. Taking the manifold graph regularizer, for example, two corrupted entries may be mis-linked and cause improper weight in a graph, and the graph regularization will mislead the unmixing and deteriorate the performance of the methods.

In this article, a NFGRM is advocated to address the aforementioned issues. In the proposed NFGRM, a sparse matrix is exploited to capture the noise-free HD. In terms of the imaging mechanism of the HD, the EV, and nonlinearity vary pixel-to-pixel and are two data-dependent interferences to the BHSU [6], [15], [23]. From the mathematical perspective, above-mentioned data-dependent interferences can incur heteroscedastic noises [25], thereupon our proposed NFGRM addresses the two interferences as data-dependent noises, and the duality and interdependence between the pixels and the endmember signatures are employed to construct reliable graph regularization. In addition, data-dependent matrix is advocated to control the ill-posedness of BHSU and the heteroscedastic noise simultaneously. The main contributions of our proposed model are summarized as follows.

1) The interferences to the BHSU problem and the noise in the HD are classified into two kinds of noises, namely data-dependent noise and data-independent noise, which are considered and addressed by a novel perspective.
2) The proposed NFGRM explores an adaptive weighted-regularizer to mitigate the adverse effects from the EV and nonlinearity, and a NMTF-based model is constructed to tackle the BHSU problem.
3) A proportional construction error matrix is introduced to control the interferences of the additive noise and the outliers, then robust regularizer is exploited via the improved noise-free HD.
4) The ill-posedness and both the heteroscedastic and homoscedastic noises are well controlled in our model.
5) Particularly, an initializing method is advocated via synthesizing the theory of the weighted LASSO estimator and the method of discriminative subspace clustering, as a result, the proposed NFGRM is insensitive to the initializations. The experimental results validate the effectiveness of the proposed NFGRM.

The rest of this article is organized as follows. Section II introduces a classical BHSU method. In Section III, the advocated model is illustrated in detail, and the corresponding algorithm is presented. In Section IV, a large number of experiments are implemented to validate the performance of the advocated model and the corresponding algorithm. Finally, the conclusions of this article are given in Section V.

II. Preliminaries

Considering a collection of mixed pixels in the HD that contain information across M spectral bands, and let $\mathbf{Y} \in \Re^{M\times P}$ denote the matrix of a HD, gathering the pixels $y_j \in \Re^M (j = 1, 2, \ldots, P)$ in its columns, where M and P are the number of spectral bands and pixels in the HD. The mixed pixels in matrix $\mathbf{Y}$ can be modeled via the widely utilized LMM, that is

$$\mathbf{Y} = \mathbf{SU} + \mathbf{E} \tag{1}$$

[1]Outlier is referred to any data point whose signature is significantly different from its surrounding background [14].

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where $\mathbf{S} = [s_1^{\mathrm{T}}, s_2^{\mathrm{T}}, \ldots, s_D^{\mathrm{T}}]^{\mathrm{T}} \in \Re^{M\times D}$ denotes the endmember matrix that contains D distinct endmember signatures, correspondingly, $\mathbf{U}\in\Re^{D\times P}$ is the abundance matrix, and $\mathbf{E} \in \Re^{M\times P}$ denotes the reconstruction error matrix (which is usually supposed to be Gaussian distributed additive noise [22], [23]). From the perspective of the physical meaning, two constraints for the abundance matrix should be enforced, i.e., the entries of $\mathbf{U}$ are nonnegative, and the sum of each column of $\mathbf{U}$ equals to 1. Moreover, as the entries of $\mathbf{S}$ and $\mathbf{E}$ represent the reflectance values and the noises, they will also be nonnegative.

The target of the BHSU is to simultaneously capture the endmember matrix and abundance matrix under the aforesaid constraints. Since NMF can factorize a matrix into two nonnegative low-rank matrices, which has been a popular tool to solve the BHSU [16]–[18]. However, NMF is an ill-posed issue, the solution space of standard NMF is too large to capture a desirable factorization for the BHSU, which implies that additional constraints should be enforced to the standard NMF. To pursue a satisfactory solution, recent works have paid much attention to exploring the prior knowledge upon the endmember signatures and abundance maps. Considering the intrinsic manifold structure and the sparsity of the HD [26], the cost function of the NMF-based BHSU can be written as

$$\begin{aligned}&\min_{\mathbf{S},\mathbf{U}} \frac{1}{2}\|\mathbf{Y}-\mathbf{S}\mathbf{U}\|_F^2 + \alpha_1\|\mathbf{U}\|_{\frac{1}{2}} + \frac{1}{2}\alpha_2 \mathrm{Tr}(\mathbf{U}\mathbf{L}\mathbf{U}^{\mathrm{T}})\\ &s.t.\ s_{id}\geq 0, \quad u_{dj}\geq 0, \ \sum_{d=1}^{D} u_{dj} = 1\\ &\qquad 1\leq i\leq M, \quad 1\leq j\leq P, 1\leq d\leq D \end{aligned} \tag{2}$$

where α_1 and α_2 are the positive tradeoff parameters, $\mathbf{L}$ is a Laplacian matrix, which originates from a nearest neighbor graph. To construct the graph for calculating the Laplacian matrix $\mathbf{L}$, let $y_i, i = 1, 2, \ldots, P$ represent a data point in M-dimension space, and assume y_i is a k-nearest neighbor of y_j; then, according to [26], the weight $\mathbf{W}_{ij}$ can be calculated via the following widely used heat kernel function, denoted as:

$$\mathbf{W}_{ij} = e^{-\frac{\|y_i - y_j\|^2}{\tau}} \tag{3}$$

where τ is the heat kernel parameter. Accordingly, the Laplacian matrix is calculated by $\mathbf{L} = \mathbf{G} - \mathbf{W}$, and $\mathbf{G}$ is a diagonal matrix calculated by $\mathbf{G}_{ii} = \sum_j \mathbf{W}_{ij}$. It is obvious that when the HD matrix $\mathbf{Y}$ is contaminated, $\mathbf{W}_{ij}$ is corrupted relatively, and their new representations u_i and u_j in the new space will be affected too. Additionally, in (2), $\|\cdot\|_F$ denotes the Frobenius norm, and $\|\mathbf{U}\|_{(1/2)}$ is defined as

$$\|\mathbf{U}\|_{\frac{1}{2}} = \sum_{d=1}^{D}\sum_{j=1}^{P} u_{dj}^{\frac{1}{2}}. \tag{4}$$

To find a local minimum of the function of (2), usually, SED is employed to measure the dissimilarity between the HD matrix $\mathbf{Y}$ and the constructed matrix $\mathbf{SU}$ [26]. Although the sparsity from abundance side and the intrinsic geometrical structure of the HD have been successfully applied in (2), the solution of (2) still can be easily affected because of the various interferences, such as, EV, additive noise, and outliers. Among them, the first one generally varies pixel to pixel [23], while the latter two ones generally seem as sparse noises [27]. Moreover, it has been demonstrated that SED is susceptible to the outliers and non-Gaussian noise [20], even a single abrupt point can arbitrarily deteriorate the quality of the construction, and what is worse is that the construction of the manifold graph heavily relies on the matrix of the HD. Due to the EV and nonlinearity, the data-dependent noises would emerge that may cause an inappropriate manifold Laplacian matrix, and then an unreliable sparsity constrained-regularization for (2). Consequently, the unmixing results will be deteriorated because of the improper regularized terms. In summary, it can be inferred that the NMF-based approach via the SED measure is unsuitable to deal with the contaminated HD.

III. Our Proposed Model and Algorithm

An increasing attention has been paid to tackle the EV, nonlinearity and outliers. These interferences are three challenging thorny impediments to the BHSU, which can incur heteroscedastic data-dependent noises and homoscedastic data-independent noises to strengthen the ill-posedness of the BHSU [6], [15], [23], [27]. Motivated by the physical mechanisms of the three interferences, we fit them into two main categories of noises: data-dependent noises and data-independent sparse noises, then the challenges are sidestepped via the advocated noise-free model.

A. On Data-Dependent Noises

Usually, the nonlinearity of a HD is considered to be incurred due to many complicated physical processes, such as intimate mixing and multiple scattering [6]. While the EV can occur in the scene of the geometry and topography of the inherent variability of the substance, or even the variation of a certain parameter (such as concentration of chlorophyll in vegetation) [23]. According to the aforementioned analyses in Section I, both EV and nonlinearity hamper the improvement of the unmixing methods' performance. Considering these two interferences vary pixel-to-pixel, we tackle EV and nonlinearity with a brand-new angle of view, that is, treating the EV and nonlinearity as the data-dependent noises.

Note that the EV generates heterogeneous structure and it relies on the unknown endmember signature to be estimated, which can bring about heteroscedastic noise and then strengthens the ill-posedness. While the weighted LASSO estimator is proven to be adaptable for eliminating the heteroscedastic noise [25], hence a variable is introduced as a data-dependent matrix to exploit the advantage of the weighted LASSO estimator. In doing so, the random fluctuations and the ill-posedness caused by $\mathbf{S}$ are mitigated, and EV is tackled. With regard to the nonlinearity, motivated by the coclustering method [20], [21], a sparser reconstruction matrix is pursued by imposing dual manifold graph constraints on the spectral side and spatial side to cocluster pixel points and features simultaneously. Note that the existing literature commonly employ the manifold regularization to exploit the spatial structure [26], [28], [29], but the spectral similarity is not considered, as the noisy bands are usually removed

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manually before unmixing. However, the preprocessing is not sufficient to perfectly remove all the corrupted points, we exploit the smoothness of the endmember signatures by enforcing the manifold regularization on the spectral side, and the deteriorated iteration because of the badly noisy bands can be avoided. Then the BHSU problem is formulated via a sparsity constrained-model, written as

$$\begin{aligned}\min_{\mathbf{S},\mathbf{U}} &\frac{1}{2}\|\mathbf{Y}-\mathbf{S}\mathbf{U}\|_F^2+\beta_1\|\mathbf{U}\|_{\frac{1}{2}}\\ &+\frac{1}{2}\beta_2\mathrm{Tr}(\mathbf{U}\mathbf{L}_\mathbf{U}\mathbf{U}^T)+\frac{1}{2}\beta_3\mathrm{Tr}(\mathbf{S}^\mathrm{T}\mathbf{L}_\mathbf{S}\mathbf{S})\\ \text{s.t. } & s_{id}\geq 0,\quad u_{dj}\geq 0,\quad \sum_{d=1}^{D}u_{dj}=1\\ & 1\leq i\leq M,\quad 1\leq j\leq P,\ 1\leq d\leq D\end{aligned}\tag{5}$$

where β_1, β_2, β_3 are small nonnegative constants to control the effects of the constrained-terms. $\mathbf{L_S}$ and $\mathbf{L_U}$ are Laplacian matrices on the spectral side graph and pixel side graph, which are, respectively, calculated from the spectral signature (row) side and pixel (column) side of the HD matrix to pursue further intrinsic structure of the HD image and provide sparser reconstruction matrix, and $\mathbf{L_S}=\mathbf{G_S}-\mathbf{W_S}$, $\mathbf{L_U}=\mathbf{G_U}-\mathbf{W_U}$, and the weight matrices $\mathbf{W_S}$ and $\mathbf{W_U}$ can be calculated by (3), accordingly, $\mathbf{G_S}$ and $\mathbf{G_U}$ are diagonal matrices of $\mathbf{W_S}$ and $\mathbf{W_U}$, written as $\mathbf{G_S}_{ii}=\sum_j \mathbf{W_S}_{ij}$ and $\mathbf{G_U}_{ii}=\sum_j \mathbf{W_U}_{ij}$. The model (5) tackles nonlinearity via introducing the sparse regularizes. It is worth mentioning that (5) neglects the cross-interactions of the EV and nonlinearity, in other words, $\beta_1\|\mathbf{U}\|_{(1/2)}$ is still corrupted because of nonlinearity, and $(1/2)\beta_2\mathrm{Tr}(\mathbf{U}\mathbf{L}_\mathbf{U}\mathbf{U}^\mathrm{T})+(1/2)\beta_3\mathrm{Tr}(\mathbf{S}^\mathrm{T}\mathbf{L}_\mathbf{S}\mathbf{S})$ can be contaminated because of the interference of the EV as well. Inspired by the theory of the weighted LASSO estimator [30], $\min_{\mathbf{S},\mathbf{U}}(1/2)\|\mathbf{Y}-\mathbf{S}\mathbf{U}\|_F^2+\beta_1\|\mathbf{U}\|_{(1/2)}$ can be represented as: $\min_{\mathbf{S},\mathbf{V}^{-1},\mathbf{X}}(1/2)\|\mathbf{Y}-\mathbf{S}\mathbf{V}^{-1}\mathbf{X}\|_F^2+\beta_1\|\mathbf{X}\|_{(1/2)}$, where $\mathbf{U}=\mathbf{V}^{-1}\mathbf{X}$, $\mathbf{X}\in\Re^{D\times P}$ is pixel side fractional matrix. $\mathbf{V}\in\Re^{D\times D}$ is a data-dependent diagonal matrix to mitigate the interference of the EV with the dth diagonal entry equal to $v_d(d=1,2,\ldots,D)$. In the light of the support of the matrices $\mathbf{S}$ and $\mathbf{X}$, the matrix $\mathbf{V}$ allows for "adaptability" with regard to the local corrupted data points. Considering the characteristics of the EV, the entries of $\mathbf{V}$ are constrained to be nonnegative. Therefore, the model (5) can be transformed into the following model:

$$\begin{aligned}\min_{\mathbf{S},\mathbf{X},\mathbf{V}^{-1}} &\frac{1}{2}\|\mathbf{Y}-\mathbf{S}\mathbf{V}^{-1}\mathbf{X}\|_F^2+\beta_1\|\mathbf{X}\|_{\frac{1}{2}}\\ &+\frac{1}{2}\beta_2\mathrm{Tr}(\mathbf{X}\mathbf{L}_\mathbf{X}\mathbf{X}^\mathrm{T})+\frac{1}{2}\beta_3\mathrm{Tr}(\mathbf{S}^\mathrm{T}\mathbf{L}_\mathbf{S}\mathbf{S})\\ \text{s.t. } & \mathbf{U}=\mathbf{V}^{-1}\mathbf{X},\quad s_{id}\geq 0,\ v_{dd}\geq 0,\ u_{dj}\geq 0,\sum_{d=1}^{D}u_{dj}=1\\ & 1\leq i\leq M,\quad 1\leq j\leq P,\ 1\leq d\leq D\end{aligned}\tag{6}$$

here, $\mathbf{L_X}$ denotes the Laplacian matrix for $\mathbf{X}$, which can be calculated by the method of calculating $\mathbf{L_U}$. And the diagonal matrix $\mathbf{G_X}$, and the weight matrix $\mathbf{W_X}$ can also be calculated as the same way of $\mathbf{G_U}$, $\mathbf{W_U}$. $\mathbf{V}^{-1}\in\Re^{D\times D}$ is the inverse of the matrix $\mathbf{V}$, and reflects the relationship between the endmember signatures and the pixel side partition matrix. According to the physical meaning of the matrix $\mathbf{V}^{-1}$, it is also a nonnegative matrix. It is obvious that the constraints of $\beta_1\|\mathbf{X}\|_{(1/2)}$ and $(1/2)\beta_2\mathrm{Tr}(\mathbf{X}\mathbf{L}_\mathbf{X}\mathbf{X}^\mathrm{T})$ in the model (6) are achieved without the support of the abundance matrix $\mathbf{U}$ but with the support of the matrix $\mathbf{X}$. Since the influence of EV is expected to be eliminated by introducing matrix $\mathbf{V}$, and nonlinearity is overcome via imposing dual manifold graph constraints on EV-free matrices $\mathbf{S}$ and $\mathbf{X}$, such that $\mathbf{S}$ and $\mathbf{X}$ are immune to the data-dependent noise, thus, the cross-interferences caused because of the EV and nonlinearity can be mitigated by avoiding constraining the abundance matrix $\mathbf{U}$ directly.

From another viewpoint, the model (5) achieves by pixel side clustering but neglects the cluster structure on the spectral side, whose framework is NMF. While NMTF has been proven to be effective for coclustering technology [20], which can group homologous data points into clusters via exploiting the inherent duality between the features and data points of a data matrix. Thus, the model (5) is extended to NMTF. In contrast to NMF, NMTF possesses the following properties: Duality and interdependence between the pixels and the endmember signatures can be exploited; Both row and column dimensions are partitioned simultaneously by constructing the relationship between spectra and pixels. Since the model (6) enjoys both the advantages of the weighted LASSO estimator and the NMTF, the variable $\mathbf{V}^{-1}$ can absorb the different scales of matrices $\mathbf{Y}$, $\mathbf{S}$, and $\mathbf{X}$, and the data-dependent noise will be eliminated as much as possible.

B. On Data-Independent Sparse Noises

In the BHSU, it is well known that the additive noise is sparse and data-independent [15], which is homoscedastic and considered as reconstruction error. Moreover, since the outliers can influence the performance of the unmixing methods with ease, the researchers have paid much attention to eliminating the interference of the outliers. Generally, the outliers consist of high-magnitude corruptions and rare endmembers, both of which are remarkably different from the background, however, the former one can seriously contaminate the learned basis matrix, the latter one should be extracted by the unmixing method. Fortunately, it has been demonstrated that the abnormal noise distributed in HD is sparse [31], while the spectrum of each substance is ideally unique, that is, the rare endmember signatures and the other ordinary endmembers are considered to distribute in a low-rank space [27]. Therefore, it can be summarized that both the additive noise and the abnormal noise are data-independent noises and locate in a sparse space.

To eliminate the interference of these noises and avoid missing the rare endmembers, low rank and sparsity should be, respectively, constrained to the rare endmembers and the noises. Since the intrinsic property of the NMF-based framework is to provide a lower rank construction of a matrix [32], hence, the model (5) can obtain the low-rank structure of the endmember matrix $\mathbf{S}$ automatically. The low-rank property is equivalently transmitted to the model (6), which guarantees the full extraction of the rare and ordinary

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endmembers. On the other hand, the reconstruction error with data-independent noise should be considered, and a sparse constraint is exploited to guarantee the sparsity of the data-independent noise. Moreover, to eliminate the interference of the abrupt noise, the quality of the gross HD matrix can be improved by subtracting partial reconstruction error or several times of the reconstruction error matrix, as a result, an approximate "data-independent noise-free" matrix is obtained by $\mathbf{Y}-\eta\mathbf{E}$, where η is a constant to control the proportion of the sparse anomalies in the construction error matrix $\mathbf{E}$.[2] Thus, combining with the model (6), considering all the challenging interferences, the problem of BHSU can be reformulated as

$$\begin{aligned}&\min_{\mathbf{S},\mathbf{X},\mathbf{V}^{-1},\mathbf{E}} \frac{1}{2}\|\mathbf{Y}-\mathbf{S}\mathbf{V}^{-1}\mathbf{X}-\eta\mathbf{E}\|_F^2+\beta_1\|\mathbf{X}\|_{\frac{1}{2}}\\&+\frac{1}{2}\beta_2\mathrm{Tr}(\mathbf{X}\mathbf{L}_\mathbf{X}\mathbf{X}^\mathrm{T})+\frac{1}{2}\beta_3\mathrm{Tr}(\mathbf{S}^\mathrm{T}\mathbf{L}_\mathbf{S}\mathbf{S})+\beta_4\|\mathbf{E}\|_{\frac{1}{2}}\\&\text{s.t. }\mathbf{U}=\mathbf{V}^{-1}\mathbf{X}, s_{id}\geq 0, v_{dd}\geq 0, u_{dj}\geq 0, \sum_{d=1}^{D}u_{dj}=1\\&\quad e_{ij}\geq 0, 1\leq i\leq M, 1\leq j\leq P, 1\leq d\leq D\end{aligned}\tag{7}$$

where β_4 is a small parameter for controlling the sparse regularizer $\|\mathbf{E}\|_{\frac{1}{2}}$. Due to the exploiting of $\eta\mathbf{E}$, a much high-quality HD matrix can be recovered from the noisy space, and more faithful unmixing result could be obtained.

C. Overall Algorithm

Evidently, in the model (7), to minimize $\mathbf{S}$, $\mathbf{V}$, $\mathbf{E}$ and $\mathbf{X}$ together is not a convex problem. Referring to the optimizing strategy of the NMF and NMTF [20], [26], [33], [34], the variables in (7) can be updated alternately, that is, when one variable is iterating, the others are fixed as constants. Consequently, each variable is iterated conditionally upon the current fixed parameters and such that the loss function is convergent. And we provide justifications in Appendix A.

Since all the parameters should be nonnegative, let the nonnegative matrices $\Upsilon\in\Re^{M\times D}$, $\Phi\in\Re^{D\times D}$, $\Psi\in\Re^{D\times P}$ and $\Theta\in\Re^{M\times P}$ be the corresponding Lagrange multipliers of the parameters $\mathbf{S}$, $\mathbf{Z}$ ($\mathbf{Z}$ is defined as a dummy variable, and $\mathbf{Z}=\mathbf{V}^{-1}$), $\mathbf{X}$ and $\mathbf{E}$. Define the Lagrange Æ as

$$\begin{aligned}Æ=&\frac{1}{2}\|\mathbf{Y}-\mathbf{S}\mathbf{Z}\mathbf{X}-\eta\mathbf{E}\|_F^2+\beta_1\|\mathbf{X}\|_{\frac{1}{2}}\\&+\frac{1}{2}\beta_2\mathrm{Tr}(\mathbf{X}\mathbf{L}_\mathbf{X}\mathbf{X}^\mathrm{T})+\frac{1}{2}\beta_3\mathrm{Tr}(\mathbf{S}^\mathrm{T}\mathbf{L}_\mathbf{S}\mathbf{S})+\beta_4\|\mathbf{E}\|_{\frac{1}{2}}\\&+\mathrm{Tr}(\Upsilon\mathbf{S}^\mathrm{T})+\mathrm{Tr}(\Phi\mathbf{Z}^\mathrm{T})+\mathrm{Tr}(\Psi\mathbf{X}^\mathrm{T})+\mathrm{Tr}(\Theta\mathbf{E}^\mathrm{T}).\end{aligned}\tag{8}$$

[2]Due to the complicated physical imaging mechanism of the HD and ill-posedness problem of the BHSU, the construction error may not only contain the sparse anomalies, but also the other entries. Moreover, in some too noisy data, the noise maybe mixed to the constructed components, in this case, the construction error is not adequate to cover the noise. We set $0<\eta\leq 2$ in this article. Note that too large value of η leads to overestimated noise and will be at the risk of nonconvergence, generally, to guarantee convergence, the value of η should not exceed 4.

Then, the partial derivative with regard to each parameter can be deduced as

$$\begin{aligned}\frac{\partial Æ}{\partial\mathbf{S}}=&-\mathbf{Y}\mathbf{X}^\mathrm{T}\mathbf{Z}^\mathrm{T}-\beta_3\mathbf{W}_\mathbf{S}\mathbf{S}+\mathbf{S}\mathbf{Z}\mathbf{X}\mathbf{X}^\mathrm{T}\mathbf{Z}^\mathrm{T}\\&+\eta\mathbf{E}\mathbf{X}^\mathrm{T}\mathbf{Z}^\mathrm{T}+\beta_3\mathbf{G}_\mathbf{S}\mathbf{S}+\Upsilon\end{aligned}\tag{9}$$

$$\frac{\partial Æ}{\partial\mathbf{Z}}=-\mathbf{S}^\mathrm{T}\mathbf{Y}\mathbf{X}^\mathrm{T}+\mathbf{S}^\mathrm{T}\mathbf{S}\mathbf{Z}\mathbf{X}\mathbf{X}^\mathrm{T}+\eta\mathbf{S}^\mathrm{T}\mathbf{E}\mathbf{X}^\mathrm{T}+\Phi\tag{10}$$

$$\begin{aligned}\frac{\partial Æ}{\partial\mathbf{X}}=&-\mathbf{Z}^\mathrm{T}\mathbf{W}^\mathrm{T}\mathbf{Y}-\beta_2\mathbf{X}\mathbf{W}_\mathbf{X}+\frac{1}{2}\beta_1\mathbf{X}^{\cdot-\frac{1}{2}}\\&+\beta_2\mathbf{X}\mathbf{G}_\mathbf{X}+\eta\mathbf{Z}^\mathrm{T}\mathbf{S}^\mathrm{T}\mathbf{E}+\mathbf{Z}^\mathrm{T}\mathbf{S}^\mathrm{T}\mathbf{S}\mathbf{Z}\mathbf{X}+\Psi\end{aligned}\tag{11}$$

$$\frac{\partial Æ}{\partial\mathbf{E}}=-\eta\mathbf{Y}+\eta\mathbf{S}\mathbf{Z}\mathbf{X}+\eta^2\mathbf{E}+\frac{1}{2}\beta_4\mathbf{E}^{\cdot-\frac{1}{2}}+\Theta\tag{12}$$

here, $()^{\cdot-(1/2)}$ denotes elementwise power of $-(1/2)$. According to the conditions of Karush–Kuhn–Tucker, the following equations are valid

$$\begin{aligned}&(-\mathbf{Y}\mathbf{X}^\mathrm{T}\mathbf{Z}^\mathrm{T}-\beta_3\mathbf{W}_\mathbf{S}\mathbf{S}+\mathbf{S}\mathbf{Z}\mathbf{X}\mathbf{X}^\mathrm{T}\mathbf{Z}^\mathrm{T}\\&\qquad+\eta\mathbf{E}\mathbf{X}^\mathrm{T}\mathbf{Z}^\mathrm{T}+\beta_3\mathbf{G}_\mathbf{S}\mathbf{S})\odot\mathbf{S}=0\\&(-\mathbf{S}^\mathrm{T}\mathbf{Y}\mathbf{X}^\mathrm{T}+\mathbf{S}^\mathrm{T}\mathbf{S}\mathbf{Z}\mathbf{X}\mathbf{X}^\mathrm{T}+\eta\mathbf{S}^\mathrm{T}\mathbf{E}\mathbf{X}^\mathrm{T})\odot\mathbf{Z}=0\\&\left(-\mathbf{Z}^\mathrm{T}\mathbf{W}^\mathrm{T}\mathbf{Y}-\beta_2\mathbf{X}\mathbf{W}_\mathbf{X}+\frac{1}{2}\beta_1\mathbf{X}^{\cdot-\frac{1}{2}}\right.\\&\qquad\left.+\beta_2\mathbf{X}\mathbf{G}_\mathbf{X}+\eta\mathbf{Z}^\mathrm{T}\mathbf{S}^\mathrm{T}\mathbf{E}+\mathbf{Z}^\mathrm{T}\mathbf{S}^\mathrm{T}\mathbf{S}\mathbf{Z}\mathbf{X}\right)\odot\mathbf{X}=0\\&\left(-\eta\mathbf{Y}+\eta\mathbf{S}\mathbf{Z}\mathbf{X}+\eta^2\mathbf{E}+\frac{1}{2}\beta_4\mathbf{E}^{\cdot-\frac{1}{2}}\right)\odot\mathbf{E}=0.\end{aligned}$$

Finally, the updating rules for each parameter can be deduced as follows:

$$\begin{aligned}\mathbf{S}=\mathbf{S}\odot\big[&(\mathbf{Y}\mathbf{X}^T\mathbf{Z}^T+\beta_3\mathbf{W}_S\mathbf{S})\oslash(\mathbf{S}\mathbf{Z}\mathbf{X}\mathbf{X}^T\mathbf{Z}^T\\&+\eta\mathbf{E}\mathbf{X}^T\mathbf{Z}^T+\beta_3\mathbf{G}_S\mathbf{S})\big]\end{aligned}\tag{13}$$

$$\mathbf{Z}=\mathbf{Z}\odot[(\mathbf{S}^T\mathbf{Y}\mathbf{X}^T)\oslash(\mathbf{S}^T\mathbf{S}\mathbf{Z}\mathbf{X}\mathbf{X}^T+\eta\mathbf{S}^T\mathbf{E}\mathbf{X}^T)]\tag{14}$$

$$\begin{aligned}\mathbf{X}=\mathbf{X}\odot\Bigg[&(\mathbf{Z}^T\mathbf{S}^T\mathbf{Y}+\beta_2\mathbf{X}\mathbf{W}_X)\\&\oslash\left(\frac{1}{2}\beta_1\mathbf{X}^{\cdot-\frac{1}{2}}+\beta_2\mathbf{X}\mathbf{G}_X\right.\\&\left.+\eta\mathbf{Z}^T\mathbf{S}^T\mathbf{E}+\mathbf{Z}^T\mathbf{S}^T\mathbf{S}\mathbf{Z}\mathbf{X}\right)\Bigg]\end{aligned}\tag{15}$$

$$\mathbf{E}=\mathbf{E}\odot\left[(\eta\mathbf{Y})\oslash\left(\eta\mathbf{S}\mathbf{Z}\mathbf{X}+\eta^2\mathbf{E}+\frac{1}{2}\beta_4\mathbf{E}^{\cdot-\frac{1}{2}}\right)\right].\tag{16}$$

1) Procedure for Overall Method: The concrete procedure is illustrated in Algorithm 1, and the involved technologies and justifications are stated as follows.

1) First, the number of endmembers is set via two available methods, namely, HySime [35] and virtual dimensionality [36]. All of the regularizer parameters and the kernel parameter τ are set between 0 and 1. The number of nearest neighbors is set to 5–10 generally. Concerning the initial values of $\mathbf{S}$, without loss of generality, the method of random initialization can be utilized, and all the elements of $\mathbf{S}$ are allocated between 0 and 1 arbitrarily, then the SCLSU [37] is employed to acquire the initial values for $\mathbf{U}$, accordingly, the construction error $\mathbf{E}$ is initialized by $\mathbf{E}=\mathbf{Y}-\mathbf{S}\mathbf{U}$. In terms of $\mathbf{Z}$, since it is a diagonal matrix, after the iteration via (14), the diagonal

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Algorithm 1 Procedure

Step 1: Input HD matrix $\mathbf{Y}$. Initialize $\mathbf{S}$, $\mathbf{U}$, $\mathbf{Z}$, $\mathbf{X}$, and $\mathbf{E}$, set the number of endmembers D and nearest neighbors, and set parameters η, β_1, β_2, β_3, β_4, β_5 and the kernel parameter τ.

Step 2: Calculate the initial spectral and pixel graph weighted matrices on the matrix $\mathbf{Y} - \eta\mathbf{E}$ with (3).

Step 3: Repeat

while *convergence condition is not satisfied* **do**

Update $\mathbf{S}$ with (13).

Update $\mathbf{Z}$ with (14), get diag($\mathbf{Z}$).

Update $\mathbf{X}$ with (15).

Update $\mathbf{E}$ with (16).

Update the spectral and pixel graph weight matrices of the matrix $\mathbf{Y} - \eta\mathbf{E}$ with (3).

Update $\mathbf{U}$ via $\mathbf{U} = \mathbf{ZX}$, and rescale each column of $\mathbf{U}$ to unit norm.

end

Step 4: Output $\mathbf{S}$, $\mathbf{U}$ and $\mathbf{E}$.

entries are taken to form a diagonal matrix $\mathbf{Z}$, which is employed to iterate for the other variables. Moreover, the initialization should be closely related to the role of $\mathbf{Z}$. Since $\mathbf{Z}$ is equal to $\mathbf{V}^{-1}$, it means that $\mathbf{Z}$ is an incidence matrix of $\mathbf{S}$ and $\mathbf{X}$ essentially. To avoid iterating long time and ensure the performance of the proposed method, an initializing method for $\mathbf{Z}$ is proposed to mitigate the issues of random fluctuations and the ill-posedness. Combining the theory of the weighted Lasso estimator with the method of discriminative subspace clustering, the initializing method is advocated and elaborated as follows:

$$\mathbf{Z} = \frac{\beta_2\mathbf{U}\mathbf{U}^{\mathrm{T}} + \beta_5\mathbf{U}\odot\mathbf{E}_2\mathbf{U}^{\mathrm{T}}}{\beta_2\mathbf{U}\mathbf{U}^{\mathrm{T}} + 0.06\mathbf{I} + \beta_5\mathbf{S}^{\mathrm{T}}\mathbf{S}\odot\mathbf{E}_1} \tag{17}$$

where β_5 is a constant to control the term. $((\beta_5\mathbf{U}\odot\mathbf{E}_2\mathbf{U}^{\mathrm{T}})/(\beta_5\mathbf{S}^{\mathrm{T}}\mathbf{S}\odot\mathbf{E}_1))$. $\mathbf{E}_1$ and $\mathbf{E}_2$ are standard deviations of the matrix $\mathbf{E}$, calculated as

$$\mathbf{E}_1 = \sqrt{\frac{1}{P}\sum_{i=1}^{P}\left[e_{\star i} - \left(\sum_{i=1}^{P} e_{\star i}\right)/P\right]^2}$$

$$\mathbf{E}_2 = \sqrt{\frac{1}{M}\sum_{i=1}^{M}\left[e_{i\star} - \left(\sum_{i=1}^{M} e_{i\star}\right)/M\right]^2}$$

$e_{\star i}$ and $e_{i\star}$,respectively, denote the column and row vectors, which belong to the matrix $\mathbf{E}$. Moreover, $\mathbf{I} \in \Re^{D\times D}$ is an identity matrix to avoid the entries of matrix $\mathbf{Z}$ getting much bigger. The term $((\beta_2\mathbf{U}\mathbf{U}^T)/(\beta_2\mathbf{U}\mathbf{U}^T))$ is obtained by the method of $\mathbf{U} = \mathbf{ZU}$ [38], [39], where the discriminative assignment for the pixel side matrix $\mathbf{X}$ can be achieved by capturing a certain matrix $\mathbf{Z}$. The residual term of (17) is inspired by the following inequality:

$$|\ (\mathbf{S}^{\mathrm{T}}\mathbf{E})_d\ | \leq \upsilon_d,\ 1 \leq d \leq D. \tag{18}$$

The inequality is homologous with the constraint in the Dantzig chooser [25], which provides a data-dependent strategy to select the weighted matrix for the weighted LASSO estimators. To better illustrate the roles of the matrix $\mathbf{V}$, let $\widehat{u}_d^{\mathbf{LS}}(d = 1, 2, \ldots, D)$ denote the estimated vector of the abundance matrix via the least-square measure, and $u_d(d = 1, 2, \ldots, D)$ is the ground truth abundance vector. According to (1), it is easy to obtain that

$$\mathbb{E}(\|\ \widehat{u}_v^{\mathbf{LS}} - u_v\ \|^2) = \sum_{v=1}^{D}\frac{\sigma_v^2}{\lambda_v^2} \tag{19}$$

where σ_v and λ_v denote the standard deviations of the heteroscedastic noise and eigenvalues of the endmember matrix, and $\mathbb{E}$ calculates mathematical expectation. According to the inequality (18) and (19), to optimize the estimated abundance matrix, the heteroscedastic noise and eigenvalues of the endmember matrix should be optimized, hence, the column and row standard deviations of the matrix $\mathbf{E}$ are calculated. Then the inequality (18) and (19) are transformed into $((\beta_5\mathbf{U}\odot\mathbf{E}_2\mathbf{U}^{\mathrm{T}})/(\beta_5\mathbf{S}^T\mathbf{S}\odot\mathbf{E}_1))$ to balance the stochastic fluctuations of the initialized components. And the term of $((\beta_5\mathbf{U}\odot\mathbf{E}_2\mathbf{U}^{\mathrm{T}})/(\beta_5\mathbf{S}^{\mathrm{T}}\mathbf{S}\odot\mathbf{E}_1))$ is added to the initializing scheme for matrix $\mathbf{Z}$. Afterward, $\mathbf{X}$ is initialized by $\mathbf{X} = \mathbf{Z}^{-1}\mathbf{U}$, and $\mathbf{E} = \mathbf{Y} - \mathbf{SZX}$.

2) On the basis of the aforementioned initializations, the manifold graphs are calculated on the "data-independent noise-free" matrix $\mathbf{Y} - \eta\mathbf{E}$. In particular, the weighted matrices of the manifold graphs are obtained from the column side and row side.

3) Then iterations are implemented. Particularly, to guarantee the sum-to-one condition for the abundance matrix, each column in the matrix $\mathbf{U}$ is rescaled to unit one in each iteration.

4) The stopping condition of the iterations is illustrated as below, that is, if the norm of the construction error $\mathbf{E}$ is not greater than the set tolerance successively for 10 times, the iteration finishes; otherwise, the updating processes continue until finishing the maximum number of iterations. Finally, the endmember, abundance and reconstruction error matrices of $\mathbf{S}$, $\mathbf{U}$, and $\mathbf{E}$ will be output.

2) Discussions: In the proposed NFGRM, the matrices $\mathbf{V}$ and $\eta\mathbf{E}$ are two pivotal determinants to the performance of the BHSU. From the mathematical viewpoint, the roles of matrix $\mathbf{V}$ are twofold. To control the stochastic fluctuations of the unmixed components; And compensating for the ill-posed problem caused because of the EV and the other potential interferences. Equation (19) indicates that the increasing of the errors between the estimated abundance vectors and the ground truth abundance vectors is proportional to the standard deviations of the heteroscedastic noise, thence, to improve the accuracy of the estimated abundance matrix, the interferences caused because of the EV and nonlinearity should be eliminated, which would incur the heteroscedastic noise to deteriorate the unmixing performance. From another

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point of view, in (19), the errors between the estimated abundance vectors and the ground truth abundance vectors are inversely proportional to the eigenvalues of the endmember matrix, while appropriate endmember signatures should locate in the space corresponding to the large eigenvalues. That is, the more accurate endmember matrix can promote generation of the more accurate abundance vectors.

In terms of the model (6) and theories of the LASSO and weighted LASSO [25], $\| \widehat{u}_\nu^{LS} - u_\nu \|^2$ should satisfy the following inequality:

$$\| \widehat{u}_\nu^{\mathbf{LS}} - u_\nu \|^2 \leq \frac{\zeta_{\beta_1}^2}{\delta} \sum_{\nu=1}^{D} \upsilon_\nu^2 \tag{20}$$

where ζ_{β_1} only depends on β_1, and δ is a parameter related to the eigenvalue condition of the endmember matrix $\mathbf{S}$, and δ corresponds to the smallest eigenvalue of $\mathbf{S}$. Moreover, according to (18) and (20), we can conclude that υ_ν is an upper bound on $|\ (\mathbf{S}^{\mathrm{T}}\mathbf{E})_\nu\ |$ and a decisive variable to the accuracy of the abundance matrix. And the upper bound of (20) is the order of $\sum_{\nu=1}^{D} \frac{\sigma_\nu^2}{\lambda_\nu^2}$, correspondingly, the estimated $\widehat{u}_\nu^{\mathbf{LS}}$ is reached the benchmark risk. That is, for the practical case where the $\lambda_\nu (\nu = 1, 2, \ldots, D)$ are much larger than λ_t (Here λ_t denotes the smallest eigenvalue of $\mathbf{S}$), and we can see the rate of (19) is controlled largely by the factor of λ_t. Henceforth, the ill-posedness is addressed by using the model (6) only in the support of the ground truth matrices $\mathbf{S}$ and $\mathbf{U}$.

In contrast to the traditional constant controlled sparse regularizer, both the ill-posedness in the support of abundance matrix and global ill-posedness are mitigated by (6). Thus, the selection of the matrix $\mathbf{V}$ is crucial and highly data-dependent. Since the EV can bring about heteroscedastic noise and then strengthens the ill-posedness, we tackle it with a novel approach by virtue of the abovementioned mathematical statements, where the data-dependent variable is exploited to constrain the regularizer, and an NMTF-based framework is constructed with the data-dependent estimator, then the matrix $\mathbf{Z}$ is introduced to mitigate the interference of the EV and improve the reliability of the dual graph estimators.

Moreover, to tackle the interference of the outlier, a robust NMTF-based model is constructed by getting rid of the abnormal data points in the gross HD.

IV. Experiments

To validate the performance of our advocated model and the corresponding algorithm, a series of experiments are conducted on synthetic and several widely used data sets, which are shown in Figs. 1 and 2. All the experiments are achieved with MATLAB 2018a code, which is implemented on a 2.9 GHz Intel Core central processing unit server via windows 10 and 64 GB memory.

In all the experiments, the kernel parameter τ in our proposed method is set as 0.1, and the number of the nearest neighbors is set as 10. We compare our proposed model with many representative methods, which are listed in Table I. Except for the noted parameter in Table I, all the other involved parameters in the compared methods are set the same with the corresponding referred literatures. Considering

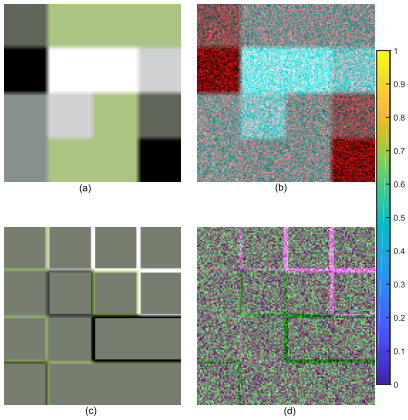


Fig. 1. Pseudocolor images of the synthetic hyperspectral data sets. The bands used in (a)–(d) as RGB channel are channels (25, 37, 59). (a) Clean image with pure pixels. (b) Noisy image with pure pixels. (c) Clean image without pure pixels. (d) Noisy image without pure pixels.

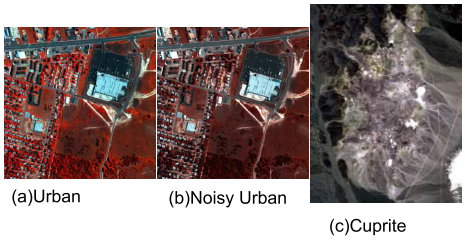


Fig. 2. Three pseudo-color images of the real hyperspectral data sets used in the experiments. The bands used in (a)–(c) as RGB channel are channels (57, 30, 20). (a) Remaining 162 bands preprocessed Urban image. (b) 210 bands original Urban image. (c) 188 bands Cuprite image.

the ill-posedness of the BHSU, every round of comparison is on the condition of a unified initializing strategy (for our proposed NFGRM, based on the unified initialized endmember and abundance matrices, the matrix $\mathbf{Z}$ is initialized by (17), and the matrix $\mathbf{X}$ is initialized by $\mathbf{X} = \mathbf{Z}^{-1}\mathbf{U}$), i.e., the endmember initializations of all the tested methods are allocated between 0 and 1 arbitrarily, and the corresponding abundance matrices are initialized by SCLSU [37].

To evaluate the proposed model NFGRM quantitatively, we employ two popularly used metrics, namely, SAD, RMSE. The formulae of SAD and RMSE are described as follows:

$$\mathrm{SAD}_\nu = \cos^{-1}\left(\frac{s_d^{\mathrm{T}}\widehat{s}_d}{\|s_d\|\,\|\widehat{s}_d\|}\right). \tag{21}$$

$$\mathrm{RMSE}_\nu = \left(\frac{1}{p}|u_d - \widehat{u}_d|^2\right)^{\frac{1}{2}} \tag{22}$$

where $\widehat{s}_d$ is the extracted endmember and its ground truth s_d, $\widehat{u}_d$, u_d are the dth estimated abundances, and real fractional abundances.

A. Experiments On Synthetic Data Set

Three synthetic data sets are utilized to evaluate the performance of proposed method NFGRM. To simulate the synthetic data sets, six reference endmembers are arbitrarily selected from the United States Geological Survey (USGS) spectral library,[3] and all the synthetic data sets are simulated with

[3]http://speclab.cr.usgs.gov/spectral.lib06

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TABLE I
COMPARED METHODS

RSU	A robust method for calculating the abundance matrix with $l_{2,1}$ norm [19].
RCMF	A constrained matrix factorization method for the BHSU with consideration of the association between the endmembers and the observed data [40] (In our experiments, sparsity level for the nonnegative subspace pursuit algorithm is set as $D+4$.).
SGSNMF	The NMF is optimized with group-structured prior information of the observed HD [41].
rNMF	An extended LMM by adding an additional term for tackling the possible nonlinear effects [42].
VR1	NMF with the logarithm of the determinant based regularizer [43].
RCONMF	Robust collaborative NMF via three steps of the unmixing chain [44].
VR2	NMF with the determinant of the Gramiant based regularizer [43].
VR3	NMF with nuclear norm regularizer [43].

224 spectral channels, each synthetic data set has 125×125 pixels, the generated abundance matrices are designed to satisfy the constraints of nonnegativity and sum-to-one. Two challenging scenarios corresponding to pure pixels existing or not are simulated with several kinds of perturbations. The simulated perturbations include additive noise, EV, outliers, and nonlinearity. The processes of the added perturbations are illustrated as follows.

1) To simulate the perturbation of EV, spectral signatures are multiplied by randomly generated scaling factors.
2) The motivation of nonlinear design is to simulate sensor-atmosphere optics. Since sensor-atmosphere optics can be treated as a GPSF [9], a bilinear mixing model is employed to simulate nonlinearity by generating a blurring kernel. And motivated by the work [9], the bilinear mixing model is written as $\mathbf{Y} = \mathbf{BSU}$. $\mathbf{B} \in \Re^{M \times M}$ is a blurring matrix, generated by 8×8 GPSF with variance 0.2.
3) Outliers are generated as follows. 10%–35% of the pixels of each synthetic data set by randomly saturating 22-130 channels, where the original reflectance values are removed and replaced by 0 or 1.
4) The additive noise is simulated to follow zero-mean white Gaussian distribution. SNR is employed to calculate the ratio of the additive noise. The expression of SNR is written as

$$\mathrm{SNR} = 10\log_{10}\frac{\mathbb{E}[\mathbf{Y}^{\mathrm{T}}\mathbf{Y}]}{\mathbb{E}[\mathbf{E}^{\mathrm{T}}\mathbf{E}]}. \tag{23}$$

1) Sensitivity Analyses to Parameters: In our proposed method NFGRM, except for the conventional kernel parameter τ, there are six dominant parameters to control the performance of the proposed method.

To investigate the sensitiveness of the parameters, a terribly contaminated synthetic data set is simulated without pure pixels, and SNR for additive noise is set to 25, scaling factor of EV is set to [0.7 1.3], outliers are added as 35% of the pixels by randomly saturating channels, and nonlinearity is added via blurring matrix **B**. To avoid the influence of the initializations, the same initializations are used in all the test experiments.

1) First, the influence of η is evaluated. The other tradeoff parameters are set to $\beta_1 = 0.34$, $\beta_2 = \beta_3 = 0.1$, $\beta_4 = 0.2$, and $\beta_5 = 0.01$. The results are shown in Fig. 3(a). As shown, if $\eta = 0$, the computational complexity is smaller and the convergence time is lower, but the performance of endmember extraction and abundances estimation are very poor. With the increasing value of η, the value of SAD becomes lower. It means that a larger value of η can yield a less noisy solution, but some useful information may be missed, i.e., $\eta = 2.4$, the results of SAD and RMSE are worse than the results of $\eta = 2.2$.
2) Second, β_1 is set in the range of $[0, 1]$. According to Fig. 3(a), η is set to 2.2, and $\beta_2 = \beta_3 = 0.1, \beta_4 = 0.2$, and $\beta_5 = 0.01$. Experimentally, a larger value of β_1 tends to force a sharper and more noisy unmixing estimation. According to Fig. 3(b), $\beta_1 = 0.6$, the best results can be obtained.
3) Third, β_2 is set to change between 0 and 0.25, and the other parameters are fixed as $\eta = 2.2$, $\beta_1 = 0.6$, $\beta_3 = 0.1$, $\beta_4 = 0.2$, and $\beta_5 = 0.01$. Fig. 3(c)

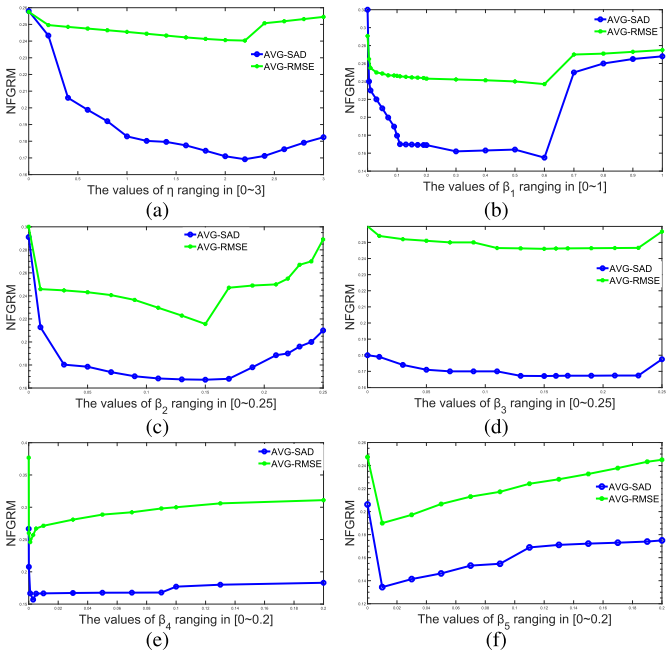


Fig. 3. Estimated results on synthetic data set with varied values of tradeoff parameters.

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TABLE II
MEAN TIME COST OF EACH RUN AND SAD AND RMSE VALUES BETWEEN THE GROUND TRUTH AND ESTIMATIONS ON THE FIRST SYNTHETIC DATA SET

	NFGRM	RSU	RCMF	SGSNMF	rNMF	VR1	RCONMF	VR2	VR3
SAD	**0.08±0.001**	\	0.28 ± 0.03	0.29 ± 0.03	0.31 ± 0.04	0.32 ± 0.03	0.37±0.04	0.19 ± 0.02	0.28 ± 0.03
RMSE $\times 10^{-5}$	**1.8±0.2**	2.7±0.4	2.7 ± 0.3	2.7 ± 0.3	2.7 ± 0.3	2.9 ± 0.4	2.3±0.3	2.5 ± 0.4	2.5±0.4
Time Cost(S)	35.9±3.2	93.6 ± 6.4	75.3 ± 5.7	68.2 ± 4.9	**10.1±1.9**	18.9 ± 3.1	10.3 ± 1.2	26.5 ± 4.2	17±1.3

shows how the performance changes with parameter β_2. The curve obviously shows that NFGRM can get the best performance when $\beta_2 = 0.15$, that is, both the values of SAD and RMSE obtain the best values.

4) Fourth, β_3 varies in the range of $[0, 0.25]$, the other parameters are set to $\eta = 2.2$, $\beta_1 = 0.6$, $\beta_2 = 0.15$, $\beta_4 = 0.2$, and $\beta_5 = 0.01$. The results are shown in Fig. 3(d). We can intuitively conclude that when $\beta_3 = 0.15$, the metric values of SAD and RMSE get minima of the curve. And the value $\beta_3 = 0$ corresponds the worst metric values, which demonstrates the effectiveness of the spectral side graph constraint.
5) Fifth, β_4 varies in the range of $[0, 0.2]$, and the other parameters are set to $\eta = 2.2$, $\beta_1 = 0.6$, $\beta_2 = \beta_3 = 0.15$, and $\beta_5 = 0.01$. As shown in Fig. 3(e), when $\beta_4 = 0.0001$, the curves for SAD and RMSE get minima.
6) Sixth, β_5 is a parameter to control the initialization of the matrix $\mathbf{Z}$, set in the range of $(0, 0.2]$, $\eta = 2.2$, $\beta_1 = 0.6$, $\beta_2 = \beta_3 = 0.15$, and $\beta_4 = 0.0001$. The results are shown in Fig. 3(f). It is obvious that $\beta_5 = 0.01$, the metric curves get the minima.

From Fig. 3(a)–(f), the parameter corresponds to the best performance in each figure is selected as a reference. Theoretically, the parameters depend on the quality of the HD, such as the spatial/spectral correlation and the density of the mixed pixels, interferences and noise. More concretely, parameter η aims at improving the quality of the HD, the choice of η relies on the noisy degree of HD, specially, if $\eta = 0$, the gross HD matrix $\mathbf{Y}$ will be unmixed directly, and the smaller value tends to capture blurry maps; β_1 controls the sparseness of the term $\mathbf{X}$, an improper value of β_1 will lead to unstable and inaccurate results. In particular, there are many endmembers in the HD and each abundance map contains multiple image patches, a smaller value will fail to extract clear and accurate abundance maps; β_3 and β_2 control the weights of spectral and pixel points graphs, respectively. Note that larger value tends to obtain a smoother endmember estimation and abundance maps. The effect of β_3 is limited due to removing the noisy bands before unmixing, however, if the corrupted bands are failed to remove, the influence of the corrupted bands can be mitigated by setting an appropriate value of β_3; the value of β_4 depends on the percentage of the data-independent noise in the HD, which can control the sparseness of the construction matrix $\mathbf{E}$. The larger value of β_4 riskily mistakes the noise as nonnoise, and blurry abundance maps will appear. Adversely, the smaller value of β_4 gives an incomplete solution. Generally, the parameter η is set in the range of $(0, 3]$ (see footnote 2); Highly mixed HD needs a bigger value of β_1; β_2 is set in the range of $(0, 0.3)$; The HD with too noisy bands needs bigger value of β_3; If the HD is terribly contaminated, a larger value of β_4 leads to provide a more noisy solution, thus, we should set a smaller value of β_4 (in particular, in the case of bigger data-independent noise but smaller data-dependent noise). Additionally, β_5 is a parameter in (17), the bigger value suits the case of highly mixed HD. Commonly, β_5 is set to $\beta_5 \approx 0.1 \times \beta_2$ or β_2. According to theoretical analysis and the empirically summarized rule, the parameters are fine-tuned in the following experiments.

Subsequently, the experiments on simulated synthetic data sets concern the following cases: Pure pixels are absent or not; The percentage of contaminated pixels varies. Fig. 1 shows two series of synthetic data sets to display the differences between clean and noisy images. Concretely, Fig.1(a) is a clean synthetic image with pure pixels but without any perturbation; Corresponding to Fig. 1(a) and (b) has pure pixels and is corrupted by aforementioned perturbations; Fig. 1(c) is a clean image without any pure pixels; In contrast to Fig. 1(c) and (d) is contaminated by several perturbations. And the images in Fig. 1(b) and (d), SNR for the additive noise is set to 30, scaling factors of EV is set as [0.9 1.1], outliers are generated as 10% of the pixels by randomly saturating 22 channels, nonlinearity is added via blurring matrix $\mathbf{B}$. It is obvious that the images Fig. 1(b) and (d) are terribly contaminated because of the interferences.

2) On Synthetic Data Set With Pure Pixels: To begin with, a synthetic data set is simulated with pure pixels, then corrupted image is employed, which is shown in Fig. 1(b). The parameters of our method NFGRM are set as follows, $\eta = 1.5$, $\beta_1 = 0.65$, $\beta_2 = 0.08$, $\beta_3 = 0.19$, $\beta_4 = 0.2$, and $\beta_5 = 0.03$. The results are shown in Fig. 4.[4] As shown, the visual difference of the extracted abundance maps is obvious among compared methods. Fig. 4 intuitively highlights the superiority of our proposed method NFGRM. Moreover, the mean values and the standard deviations of the SAD and RMSE over 20 runs are shown in Table II. The best results are highlighted via bold fonts. As can be observed in Table II, except for our method NFGRM, the other compared methods perform poor performances. Moreover, compared with the other tested methods, such as spatial group sparsity regularized NMF (SGSNMF), rNMF, and NFGRM performs better and is immune to the randomly initialization, which verifies the superiority of the proposed initializing method in (17). By adding the antinoise terms and sparse constraints, our advocated method NFGRM can further improve the performance, which explains the results of NFGRM why outperform that of the other algorithms.

[4]The proportion of the large contribution of the substance corresponds to a red pixel; On the contrary, the small contributed substance corresponds to a blue pixel. Consequently, the pure red denotes that the ratio of a certain category of material in this pixel is 1, and pure blue represents 0.

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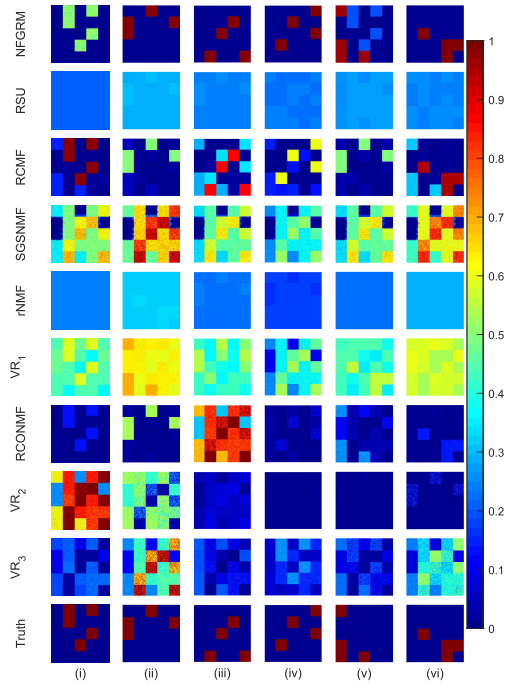


Fig. 4. Ground truth and extracted abundance maps for the synthetic data set. The color scale is 0(blue)-1(red) [4].

3) Robustness Analysis to Varied Outliers Corruption: To evaluate the robustness to outlier corruption, two series of synthetic data sets are simulated without pure pixels. In this part, the additive noise, nonlinearity, and EV are simulated the same with Fig. 1(b). In the first series of synthetic data sets, the percentage of the pixels is set as 10% for outliers, and 37, 44, and 74 channels are, respectively, saturated. In the second series of synthetic data sets, the number of saturated channels is set to 112, and the percentage of the pixels is set to $15\%, 20\%, 25\%, and 30\%$, respectively. The tradeoff parameters of our method NFGRM are set as follows, $\eta = 2.2$, $\beta_1 = 0.6$, $\beta_2 = 0.1$, $\beta_3 = 0.15$, $\beta_4 = 0.002$, and $\beta_5 = 0.01$. A series of comparisons using the heavily mixed data sets are shown in Fig. 5, where average (AVG)-SAD and AVG-RMSE correspond to average values of SAD and RMSE. As shown in Fig. 5(a) and (c), due to the increasing of the outliers, all the algorithms yield poor values of SAD, however, the SAD values of our advocated method NFGRM are superior to the other methods, and the differences for the values of AVG-SAD obtained from NFGRM are approximately equal to zero, which indicates that endmember extracting performance of NFGRM is insensitive to the increasing outliers. With regard to estimating abundances, the AVG-RMSE comparison also indicates the superiority of our proposed method NFGRM. Why NFGRM can perform better than the others, the reasons are concluded as: ⟨1⟩Although random initialization is utilized, our proposed method NFGRM is insensitive to the initialization; ⟨2⟩By contrast, our method NFGRM is more robust to the perturbations.

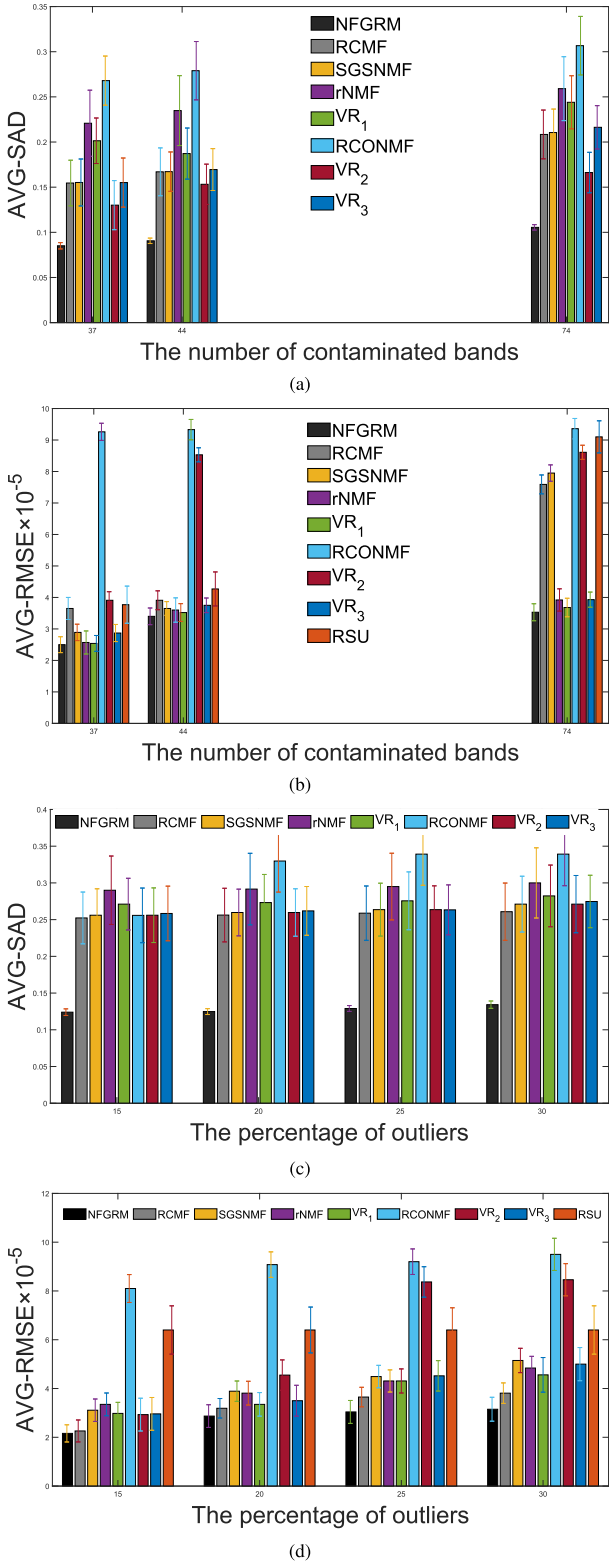


Fig. 5. Estimated results on synthetic data set with varied number of contaminated bands and varied percentage of outliers. (a) AVG-SAD with varied number of contaminated bands. (b) AVG-RMSE with varied number of contaminated bands. (c) AVG-SAD with varied percentage of outliers saturated in 112 bands. (d) AVG-RMSE with varied percentage of outliers.

B. Experiments On Urban Data Set

The Urban data were captured in October, 1995, via hyperspectral digital imagery collection experiment (HYDICE) sensors. This data set contains 210 bands, and the range of the wavelengths is $0.4 - 2.5\mu m$. With a spatial resolution of 2 m, a subimage containing 307 × 307 pixels is used in our experiments. Generally, before unmixing, the data set is preprocessed by removing the badly contaminated bands (the removed bands are 1–4, 76, 87, 101–111, 136–153, and

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TABLE III
MEAN TIME COST OF EACH RUN ON 162 AND 210 BANDS URBAN DATA SET

	NFGRM	RSU	RCMF	SGSNMF	rNMF	VR1	RCONMF	VR2	VR3
162 bands(S)	271.7±19.8	513.8 ± 33.5	590.3 ± 200.4	66.2 ± 4.9	31.2±9.8	30.3 ± 2.7	62.9 ± 6.1	30.1 ± 2.1	**30.1±2**
210 bands(S)	299.1±30.25	604 ± 20.24	480 ± 14.9	61.5 ± 4.5	**27.2±3.8**	30.16±3.55	51.6±4.22	30.26 ± 2.98	30.2±2.91

198–210), 162 bands are utilized in the unmixing. According to[5] (Note that the reference endmembers were manually selected, and the reference abundance maps have been obtained by a standard linear unmixing method, thus not accounting for all the complexity present in real data (such as EV and nonlinearity). In contrast to the reference ground truth, NFGRM is expected to simultaneously tackle EV, outlier, nonlinearity and additive noise. Moreover, unlike the reference endmembers, NFGRM is implemented with randomly initialized endmembers) and [45], there are four endmembers in the Urban data set. Moreover, to test the capability of tackling the badly degraded and complicated mixed data, both the preprocessed Urban data and original Urban data are utilized in this section, i.e., experiments are implemented, respectively, on preprocessing Urban data set with 162 bands and gross noisy urban data set with 210 bands, which are shown in Fig. 2(a) and (b). It can be observed visually that Fig. 2(b) is blurrier than Fig. 2(a).

1) On Urban Data Set: Here, 162 bands Urban data set is utilized, 20 runs of each tested method are implemented. In our proposed method, $\eta = 0.88$, $\beta_1 = 0.4$, $\beta_2 = 0.16$, $\beta_3 = 0.16$, $\beta_4 = 0.3$, and $\beta_5 = 0.02$. The results are shown in Fig. 6 (since the estimated abundance maps by robust spectral unmixing (RSU) and rNMF are too dark to see, all the entries in the corresponding abundance matrices are amplified by multiplying a constant 3.5) and Table III. From Fig. 6(b), it can be concluded visually that the estimated abundance maps via our method coincide with the reference ground truth Fig. 6(a)[5] better than the other methods. Since the computational complexity of our method contains the iterations of three nonnegative components and double manifold graph regularized terms, the time cost is higher than some compared methods, which is shown in Table III.

2) On Noisy Urban Data Set: The original Urban data set is employed to display the robustness of the proposed method NFGRM to the noise. Since the degree of corruptions in Urban 210 bands data set is higher than Urban 162 bands data set, we set $\eta = 1.8$. The other parameters of our method are set as follows, $\beta_1 = 0.4$, $\beta_2 = 0.16$, $\beta_3 = 0.19$, $\beta_4 = 0.003$, and $\beta_5 = 0.02$. The abundance maps are estimated by all the tested methods, but just our method can estimate the abundance maps well. The abundance maps estimated by the above-mentioned compared methods are poor and are hardly readable, hence, only the abundance maps estimated by our proposed method NFGRM are shown in Fig. 7, and the time costs are shown in Table III. It is obvious that the overall results on the noisy Urban data set are worse than the results on the 162 bands Urban data set. Fortunately, our proposed method captures the most patches inside each abundance map.

[5]http://lesun.weebly.com/hyperspectral-data-set.html

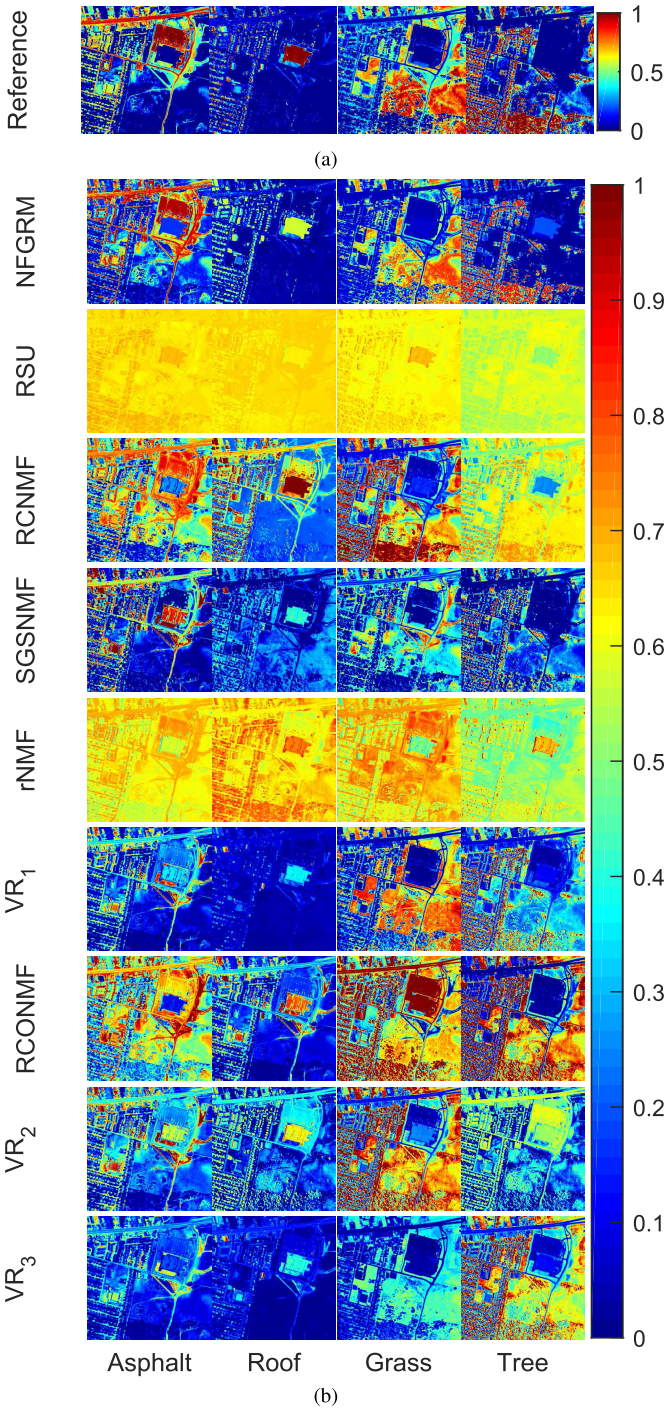


Fig. 6. Reference[5] and estimated abundance maps for the Urban image. The color scale is 0(blue)-1(red)[4].

In Fig. 7, the estimated abundance maps of asphalt, grass, tree, and roof by our proposed method well coincide with the reference ground truth. It implies that our proposed method can well estimate the abundance maps on the noisy and heavily

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TABLE IV

MEAN TIME COST OF EACH RUN AND SAD VALUES BETWEEN THE GROUND TRUTH[3] AND ESTIMATED RESULTS ON CUPRITE DATA SET

	NFGRM	RCMF	SGSNMF	rNMF	VR1	RCONMF	VR2	VR3
a	**0.0164±0.003**	0.0401 ± 0.005	0.5739 ± 0.068	1.5625 ± 0.571	0.9418±0.076	0.3629 ± 0.047	0.3012 ± 0.055	0.5236±0.079
b	**0.1085±0.006**	0.13 ± 0.026	0.6225 ± 0.077	1.5633 ± 0.393	0.9744±0.101	0.3017 ± 0.061	0.2948 ± 0.044	0.5785±0.091
c	**0.1479±0.037**	0.1661 ± 0.0041	0.6297 ± 0.074	1.5635 ± 0.243	0.9724±0.102	0.2919 ± 0.043	0.3033 ± 0.051	0.5904±0.069
d	**0.0831±0.021**	0.1219±0.048	0.5616 ± 0.081	1.5627±0.631	0.9098 ± 0.087	0.3691 ± 0.049	0.3249±0.056	0.4506 ± 0.064
e	0.1379±0.021	**0.1356±0.023**	0.5587±0.061	1.5627 ± 0.612	0.903±0.079	0.3738 ± 0.041	0.3316 ± 0.043	0.4395±0.056
f	**0.0957±0.006**	0.0997 ± 0.011	0.5769 ± 0.042	1.5624 ± 0.58	0.9212±0.039	0.327 ± 0.041	0.2979 ± 0.031	0.4771±0.054
g	**0.0992±0.006**	0.1185 ± 0.038	0.6028±0.078	1.5627 ± 0.354	0.9369 ± 0.083	0.2837 ± 0.045	0.2782±0.053	0.5138±0.067
h	**0.0464±0.007**	0.0572 ± 0.009	0.5781 ± 0.072	1.5628 ± 0.461	0.9406±0.078	0.3468 ± 0.029	0.3016 ± 0.045	0.5179±0.063
i	**0.1659±0.024**	0.21 ± 0.033	0.6522 ± 0.061	1.5637 ± 0.713	1.003±0.114	0.3201 ± 0.046	0.3328 ± 0.057	0.6446±0.071
j	**0.0805±0.004**	0.087 ± 0.008	0.6061 ± 0.081	1.5629 ± 0.711	0.976±0.069	0.3382 ± 0.047	0.3019 ± 0.041	0.5746±0.066
k	**0.0842±0.005**	0.0872 ± 0.009	0.5732 ± 0.076	1.5628 ± 0.298	0.9257±0.093	0.3415 ± 0.042	0.3076 ± 0.041	0.4883±0.061
l	**0.0804±0.006**	0.0897 ± 0.009	0.5916 ± 0.091	1.5628 ± 0.453	0.9408 ± 0.087	0.3091 ± 0.056	0.284±0.044	0.5241 ± 0.066
Mean	**0.0955±0.006**	0.1119 ± 0.017	0.5939 ± 0.002	1.5629 ± 0.05	0.9455±0.005	0.3305 ± 0.018	0.305±0.014	0.5269 ± 0.017
Time Cost(S)	82.2±6.7	604.3 ± 46.3	76.1 ± 6.9	37.33 ± 4.1	30.3±2.6	64.4±5.1	30.3±2.9	**30.3±2.3**

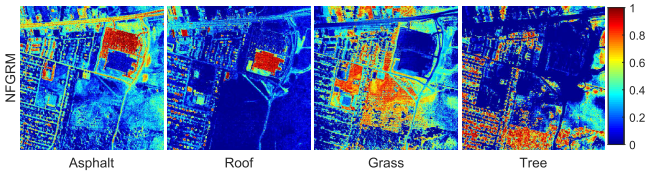


Fig. 7. Estimated fractional abundance maps for the noisy Urban data set. The color scale varies in a range of 0(blue)-1(red)[4].

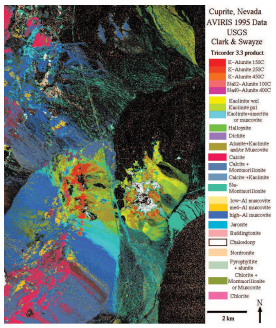


Fig. 8. Reference abundance maps for the Cuprite data set.

corrupted data. Moreover, Fig. 7 verifies that the capability of our proposed method to tackle the badly contaminated data is more outstanding than the compared methods. Note that some methods consume longer time on 162 bands data set than on 210 bands data set, this phenomenon illustrates that the random initializing strategy causes some methods to consume longer convergence time even on a smaller data set.

C. Experiments on Cuprite Data Set

The Cuprite data set is available online,[6] has been widely used in unmixing research. It was obtained by the airborne visible / infrared imaging spectrometer (AVIRIS), covering the Cuprite mining site, Nevada. The reference map of the imaging area can be found in.[7] In our experiments, a subset of $250\times191\times188$ is selected, which is shown in Fig. 2(c). And the reference abundance map is shown in Fig. 8. According to some existing research [9], [26], 12 materials utilized, and publicly identified mineral spectra were chosen from the USGS[3] library as references.

Here, $\eta = 0.8$. Considering numerous mixed minerals in the Cuprite data set, most of the parameters of our proposed

[6]https://aviris.jpl.nasa.gov/data/free_data.html

[7]https://speclab.cr.usgs.gov/cuprite95.tgif.2.2 *um_map*.gif

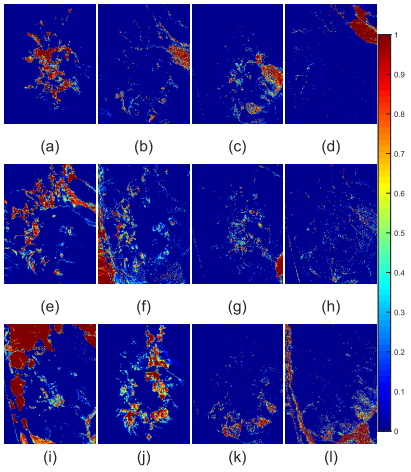


Fig. 9. Extracted abundance maps by the proposed method NFGRM on the Cuprite data set. The color scale varies from 0(blue) to 1(red)[4].

method NFGRM are set larger than on the 162 bands Urban data set, i.e., $\beta_1 = 1.1$, $\beta_2 = 0.2$, $\beta_3 = 0.16$, $\beta_4 = 1$, and $\beta_5 = 0.2$. On the condition of the random initializations, the estimated abundance maps by the methods of RSU, robust constrained matrix factorization (RCMF), SGSNMF, rNMF, robust collaborative NMF (RCONMF), VR_1, VR_2, and VR_3 contain many ambiguous and repetitive contents (see Appendix B), thus, we just show the estimated abundance maps by our proposed method NFGRM. And we show the estimated abundance maps in Fig. 9. Since the abundances of the ground truth are absent, only the SAD is used to measure the accuracy of the tested methods, and 25 runs of the SAD mean values and the standard deviations between the ground truth[3] and estimated results are illustrated. It is worth noting that RSU is a method for just calculating the abundance matrix, and except the method of RSU, the results of the other 8 methods are listed in Table IV. Moreover, the time consumption is listed in Table IV.

According to [7], Fig. 9 shows that the recovered abundance maps by our advocated method coincide with the ground truth perfectly. Particularly, for (a) Chalcedony, (g) Montmorillonite, (j) Kaolinte, and (k) Alunite. Compared to the other methods (see Appendix B), since NFGRM is immune to the interferences and employs dual graph constraints to pursue sparser reconstruction, it identifies all the 12 substances more accurately, and the extracted maps with higher pureness than the other compared methods. Moreover, in Table IV, it is obvious our proposed method NFGRM identifies the most

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endmembers. Except for the endmember (e), NFGRM obtains better SAD values than the other seven methods. In other words, in all the tested methods, our method captures the best unmixing results. Nevertheless, the high computational complexity causes long time consumption, which is longer than the methods of VR1, VR2, VR3, SGSNMF, rNMF, and RCONMF.

V. Conclusion

In this article, all kinds of the interferences are classified into two categories: data-dependent noise and data-independent noise. Under this premise, a data-dependent sparse regularizer is proposed to tackle the problem of EV in the BHSU. Then a robust NMTF-based model with dual manifold graph regularizer is constructed on the basis of the proposed scaled sparse regularized model, and noise-free HD can be captured by removing proportionate residual term from the original HD. Moreover, the advocated model exploits the manifold structures from both the spectral and spatial sides to pursuit a sparser reconstruction space. The advocated model simultaneously and synergistically tackles the problems of the EV, nonlinearity, outlier, and additive noise. The experiments are implemented on synthetic and two extensively utilized data sets with random initializations. In particular, unlike traditional research, the Urban data set with 162 bands and 210 bands are employed to demonstrate the performance of antiinterference. In all the experiments, our proposed method can identify the endmembers accurately and recover pure abundance maps, even on the original noisy 210 bands Urban data set. And results of our proposed method are better than the other tested methods.

Considering the computational complexity of our proposed method and multiple parameters, we will focus on efficient computing and develop a simplified powerful model.

Appendix A
Proof for Convergence of Proposed Method

Although it is difficult to theoretically prove the convergence of the proposed method, the convergence of each iteration of the procedure can be guaranteed, and we provide justifications for convergence as below.

If $\mathbf{S}$ is updated, $\mathbf{Z}$, $\mathbf{X}$, and $\mathbf{E}$ will be fixed as constants, the convergent problem is equivalent to the literature [34]. Similarly, to update $\mathbf{V}$ by fixing $\mathbf{S}$, $\mathbf{X}$, and $\mathbf{E}$. Since all the variables are nonnegative, the convergence will be guaranteed by the theory in the literature [46]. With regard to the presence of the $L_{1/2}$ norms in updating $\mathbf{X}$, we have referenced the literature [26], [33]. Literature [33] proved $L_{1/2}$ sparsity-constrained NMF can be convergent. And literature [26] demonstrated incorporating the graph regularization term into the $L_{1/2}-$ NMF can be convergent. In our proposed method, to update $\mathbf{X}$, the other variables will be fixed as constants, and $\mathbf{X}$ is a nonnegative matrix with graph regularization term and $L_{1/2}$ norm, the iteration for $\mathbf{X}$ is similar to updating abundance matrix in the literature [26], hence, the convergent guarantee in [26] is available for the matrix $\mathbf{X}$. Hence, we focus on the convergence of updating matrix $\mathbf{E}$. To guarantee the convergence of the iterating rule for $\mathbf{E}$. Let e as a vector of $\mathbf{E}$. An auxiliary function $\mathrm{Q}(e, e^t)$ is given for vector e, and it satisfies the conditions $\mathrm{Q}(e, e) = \mathrm{G}(e)$ and $\mathrm{Q}(e, e^t) \geq \mathrm{G}(e)$, such that $\mathrm{G}(e)$ is nonincreasing when iteration using the following equation:

$$e^{t+1} = \min_{e} \mathrm{Q}(e, e^t). \tag{24}$$

It is guaranteed by

$$\mathrm{G}(e^{t+1}) \leq \mathrm{Q}(e^{t+1}, e^t) \leq \mathrm{Q}(e^t, e^t) = \mathrm{G}(e^t). \tag{25}$$

According to reference [33], the function Q can be defined as

$$\mathrm{Q}(e, e^t) = \mathrm{G}(e^t) + (e - e^t)(\nabla \mathrm{G}(e^t))^{\mathrm{T}} + \frac{1}{2}(e - e^t)\mathbf{C}(e^t)(e - e^t)^{\mathrm{T}}. \tag{26}$$

$\mathbf{C}(e^t)$ is a diagonal matrix and defined as

$$\mathbf{C}(e^t) = \mathrm{diag}\left(\left(\eta szx + \eta^2 e^t + \frac{1}{2}\beta_4 (e^t)^{-\frac{1}{2}}\right) \oslash e^t\right). \tag{27}$$

Due to $\mathrm{Q}(e, e) = \mathrm{G}(e)$, the Taylor expansion of $\mathrm{G}(e)$ can be written as

$$\mathrm{G}(e) = \mathrm{G}(e^t) + (e - e^t)(\nabla \mathrm{G}(e^t))^T + \frac{1}{2}(e - e^t) \times \left(\eta^2 - \frac{1}{4}\beta_4 \mathrm{diag}\left((e^t)^{-\frac{3}{2}}\right)\right)(e - e^t)^{\mathrm{T}} + Ł(\mathrm{rest}). \tag{28}$$

Here, Ł(rest) represents the Lagrange remainder. And note that the constraint $\mathrm{Q}(e, e^t) \geq \mathrm{G}(e)$ is satisfied on the condition of

$$\begin{aligned}(e - e^t)&\left(\mathbf{C}(e^t) - \eta^2 + \frac{1}{4}\beta_4 \mathrm{diag}\left((e^t)^{-\frac{3}{2}}\right)\right)(e - e^t)^{\mathrm{T}} \geq 0 \\ \Rightarrow\ (e - e^t)&\left(Ç(e^t) + \frac{1}{2}\beta_4 (e^t)^{-\frac{1}{2}} + \frac{1}{4}\beta_4 \mathrm{diag}\left((e^t)^{-\frac{3}{2}}\right)\right)(e - e^t)^{\mathrm{T}} \geq 0.\end{aligned} \tag{29}$$

In expressions (29), Ł(rest) has been omitted as it emerges on both sides of the inequality. $Ç(e^t)$ is defined as

$$Ç(e^t) = \mathrm{diag}[(\eta szx + \eta^2 e^t) \oslash e^t] - \eta^2. \tag{30}$$

Lee and Seung [46] proved the positive semidefiniteness of $\mathrm{diag}((\mathbf{S}^{\mathrm{T}}\mathbf{S}u^t) \oslash u^t) - \mathbf{S}^{\mathrm{T}}\mathbf{S}$, in analogy to $Ç(e^t)$, η is a small constant, generally, set as $0 < \eta \leq 2$, and szx is the reconstructed vector for the HD image, it should be a nonnegative matrix and the energy of szx is generally larger than the reconstruction error vector e. Since larger value of η generates larger energy for error vector e, which may exceed the energy of the reconstruction vector szx, hence, to ensure convergence, $Ç(e^t)$ must be a positive semidefinite matrix, and η should not be too big. Due to the nonnegativity of e, the other two terms in (29) are nonnegative. Furthermore, (29) holds by substituting (26) into (24). This leads to the update rule

$$\begin{aligned}e^{t+1} &= e^t - \nabla \mathrm{G}(e^t) C^{-1}(e^t) \\ &= e^t - (\eta szx + \eta^2 e^t + \frac{1}{2}\beta_4 e^{t\,-\frac{1}{2}} - \eta \mathbf{Y}) \\ &\quad \odot e^t \oslash \left(\eta szx + \eta^2 e^t + \frac{1}{2}\beta_4 e^{t\,-\frac{1}{2}}\right) \\ &= e^t \odot \left((\eta \mathbf{Y}) \oslash \left(\eta szx + \eta^2 e^t + \frac{1}{2}\beta_4 (e^t)^{\frac{-1}{2}}\right)\right).\end{aligned} \tag{31}$$

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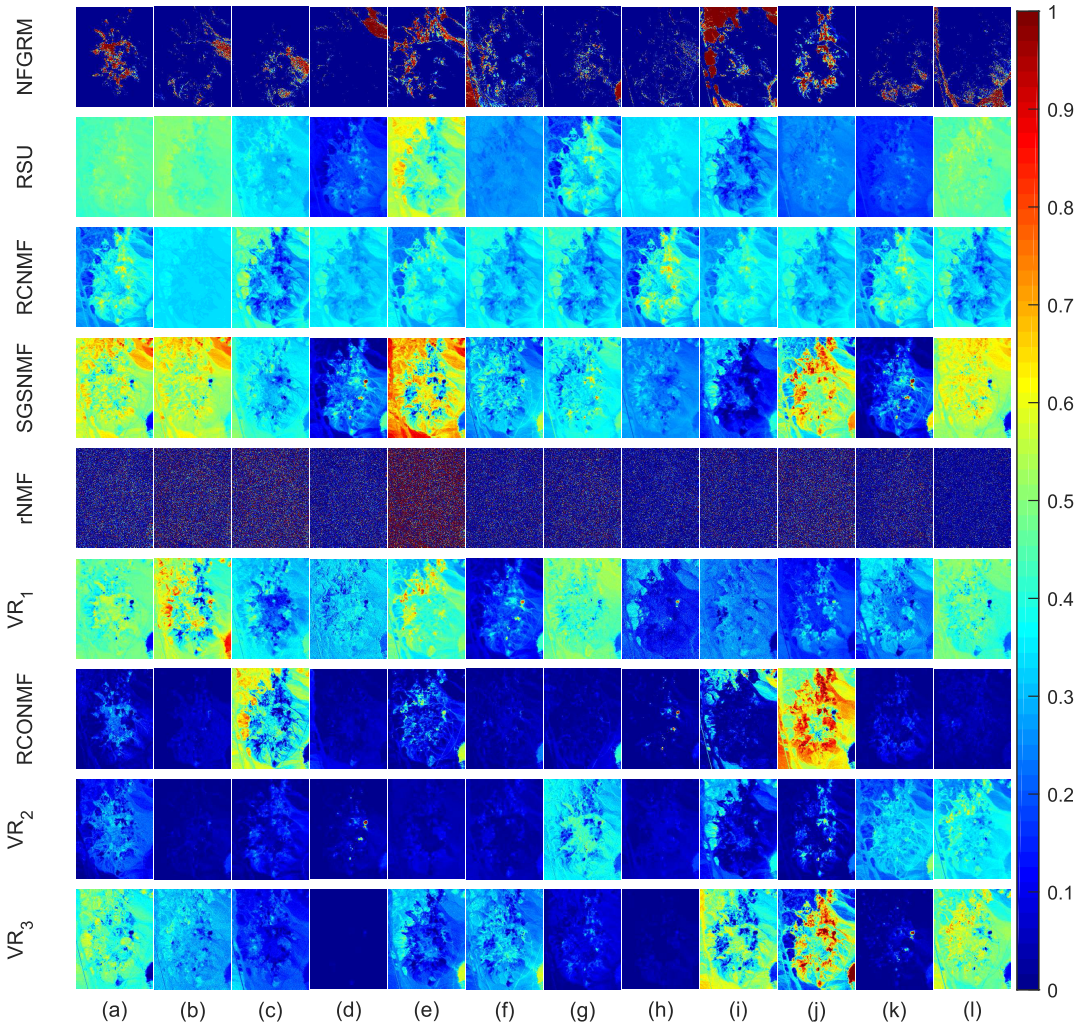


Fig. 10. Estimated abundance maps for the Cuprite data set. The color scale varies from 0(blue) to 1(red)[4].

APPENDIX B
ESTIMATED ABUNDANCES FOR CUPRITE DATA SET

We compare the estimated abundance maps by NFGRM with the other 8 methods, which are shown in Fig. 10. According to the reference[7], Fig. 10 shows that the abundance maps recovered by our advocated method coincide with the ground truth perfectly, particularly, for (a) Chalcedony, (g) Montmorillonite, (j) Kaolinte, and (k) Alunite etc. Compared to the other methods, NFGRM identifies all the 12 substances more accurately and the extracted maps with higher pureness than the other compared methods, however, the estimated abundance maps of the other methods contain many ambiguous and repetitive contents.

REFERENCES

[1] R. Arablouei and F. de Hoog, "Hyperspectral image recovery via hybrid regularization," *IEEE Trans. Image Process.*, vol. 25, no. 12, pp. 5649–5663, Dec. 2016.

[2] J. Huang, T.-Z. Huang, L.-J. Deng, and X.-L. Zhao, "Joint-sparse-blocks and low-rank representation for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2419–2438, Apr. 2019.

[3] R. A. Borsoi, T. Imbiriba, J. C. M. Bermudez, and C. Richard, "A fast multiscale spatial regularization for sparse hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 598–602, Apr. 2019.

[4] M. Wang, B. Zhang, X. Pan, and S. Yang, "Group low-rank nonnegative matrix factorization with semantic regularizer for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1022–1029, Apr. 2018.

[5] Y. E. Salehani, S. Gazor, and M. Cheriet, "Sparse hyperspectral unmixing via heuristic ℓ_p-norm approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1191–1202, Apr. 2018.

[6] J. Sigurdsson, M. O. Ulfarsson, and J. R. Sveinsson, "Blind sparse nonlinear hyperspectral unmixing using an ℓ_q penalty," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 12, pp. 1907–1911, Dec. 2018.

[7] D. Wang, Z. Shi, and X. Cui, "Robust sparse unmixing for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1348–1359, Mar. 2018.

[8] Y. Woodbridge, U. Okun, G. Elidan, and A. Wiesel, "Unmixing K-Gaussians with application to hyperspectral imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7281–7293, Sep. 2019, doi: 10.1109/TGRS.2019.2912818.

[9] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Blind hyperspectral unmixing considering the adjacency effect," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6633–6649, Sep. 2019.

[10] X. Jiang, M. Gong, H. Li, M. Zhang, and J. Li, "A two-phase multiobjective sparse unmixing approach for hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 508–523, Jan. 2018.

[11] C.-H. Lin, C.-Y. Chi, Y.-H. Wang, and T.-H. Chan, "A fast hyperplane-based minimum-volume enclosing simplex algorithm for blind hyperspectral unmixing," *IEEE Trans. Signal Process.*, vol. 64, no. 8, pp. 1946–1961, Apr. 2016.

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[12] J. Li, A. Agathos, D. Zaharie, J. M. Bioucas-Dias, A. Plaza, and X. Li, "Minimum volume simplex analysis: A fast algorithm for linear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 5067–5082, Sep. 2015.
[13] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.
[14] S. Nakhostin, H. Clenet, T. Corpetti, and N. Courty, "Joint anomaly detection and spectral unmixing for planetary hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 6879–6894, Dec. 2016.
[15] J. M. Bioucas-Dias *et al.*, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.
[16] H. Fang, A. Li, H. Xu, and T. Wang, "Sparsity-constrained deep nonnegative matrix factorization for hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 7, pp. 1105–1109, Jul. 2018.
[17] Y. E. Salehani and S. Gazor, "Smooth and sparse regularization for NMF hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3677–3692, Aug. 2017.
[18] X.-R. Feng, H.-C. Li, J. Li, Q. Du, A. Plaza, and W. J. Emery, "Hyperspectral unmixing using sparsity-constrained deep nonnegative matrix factorization with total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6245–6257, Oct. 2018.
[19] Y. Ma, C. Li, X. Mei, C. Liu, and J. Ma, "Robust sparse hyperspectral unmixing with $\ell_{2,1}$ norm," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1227–1239, Mar. 2017.
[20] L. Du and Y.-D. Shen, "Towards robust co-clustering," in *Proc. 23rd Int. Joint Conf. Artif. Intell.*, A. Publications, Ed., Jun. 2013, pp. 1317–1322.
[21] H. Wang, F. Nie, H. Huang, and F. Makedon, "Fast nonnegative matrix tri-factorization for large-scale data co-clustering," in *Proc. 22nd Int. Joint Conf. Artif. Intell.*, 2011, pp. 1553–1557.
[22] C. Li, X. Chen, and Y. Jiang, "On diverse noises in hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5388–5402, Oct. 2015.
[23] L. Drumetz, M.-A. Veganzones, S. Henrot, R. Phlypo, J. Chanussot, and C. Jutten, "Blind hyperspectral unmixing using an extended linear mixing model to address spectral variability," *IEEE Trans. Image Process.*, vol. 25, no. 8, pp. 3890–3905, Aug. 2016.
[24] L. Drumetz, T. R. Meyer, J. Chanussot, A. L. Bertozzi, and C. Jutten, "Hyperspectral image unmixing with endmember bundles and group sparsity inducing mixed norms," *IEEE Trans. Image Process.*, vol. 28, no. 7, pp. 3435–3450, Jul. 2019.
[25] X. J. Hunt, P. Reynaud-Bouret, V. Rivoirard, L. Sansonnet, and R. Willett, "A data-dependent weighted LASSO under Poisson noise," *IEEE Trans. Inf. Theory*, vol. 65, no. 3, pp. 1589–1613, Mar. 2019.
[26] X. Lu, H. Wu, Y. Yuan, P. Yan, and X. Li, "Manifold regularized sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2815–2816, Oct. 2013.
[27] Y. Qu *et al.*, "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.
[28] L. Tong, J. Zhou, Y. Qian, and Y. Gao, "Multiple graph regularized NMF for hyperspectral unmixing," in *Proc. 7th Workshop Hyperspectral Image Signal Process. Evol. Remote Sens. (WHISPERS)*, Tokyo, Japan, Jun. 2015, pp. 1–4.
[29] R. Rajabi and H. Ghassemian, "Sparsity constrained graph regularized NMF for spectral unmixing of hyperspectral data," *J. Indian Soc. Remote Sens.*, vol. 43, no. 2, pp. 269–278, Jun. 2015.
[30] S. van de Geer, P. Bühlmann, and S. Zhou, "The adaptive and the thresholded lasso for potentially misspecified models (and a lower bound for the Lasso)," *Electron. J. Statist.*, vol. 5, pp. 688–749, Jul. 2011.
[31] Y. Yuan, Q. Wang, and G. Zhu, "Fast hyperspectral anomaly detection via high-order 2-D crossing filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 620–630, Feb. 2015.
[32] D. Kuang, S. Yun, and H. Park, "SymNMF: Nonnegative low-rank approximation of a similarity matrix for graph clustering," *J. Global Optim.*, vol. 62, no. 3, pp. 545–574, Jul. 2015.
[33] Y. Qian, S. Jia, J. Zhou, and A. Robles-Kelly, "Hyperspectral unmixing via $L_{1/2}$ sparsity-constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4282–4297, Jun. 2011.
[34] F. Shang, L. C. Jiao, and F. Wang, "Graph dual regularization nonnegative matrix factorization for co-clustering," *Pattern Recognit.*, vol. 45, no. 6, pp. 2237–2250, Jun. 2012.
[35] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.
[36] C.-I. Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.
[37] M. Veganzones *et al.*, "A new extended linear mixing model to address spectral variability," in *Proc. 6th Workshop Hyperspectral Image Signal Process. Evol. Remote Sens. (WHISPERS)*, Lausanne, Switzerland, Jun. 2014, pp. 1–4.
[38] C. Lu, J. Feng, Z. Lin, T. Mei, and S. Yan, "Subspace clustering by block diagonal representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 2, pp. 487–501, Feb. 2019.
[39] R. Ammanouil, A. Ferrari, C. Richard, and D. Mary, "Blind and fully constrained unmixing of hyperspectral images," *IEEE Trans. Image Process.*, vol. 23, no. 12, pp. 5510–5518, Dec. 2014.
[40] N. Akhtar and A. Mian, "RCMF: Robust constrained matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3354–3366, Jun. 2017.
[41] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6287–6304, Nov. 2017.
[42] C. Fevotte and N. Dobigeon, "Nonlinear hyperspectral unmixing with robust nonnegative matrix factorization," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 4810–4819, Dec. 2015.
[43] A. M. S. Ang and N. Gillis, "Algorithms and comparisons of nonnegative matrix factorizations with volume regularization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4843–4853, 2019.
[44] J. Li, J. M. Bioucas-Dias, A. Plaza, and L. Liu, "Robust collaborative nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6076–6090, Oct. 2016.
[45] J. Yao, D. Meng, Q. Zhao, W. Cao, and Z. Xu, "Nonconvex-sparsity and nonlocal-smoothness-based blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2991–3006, Jun. 2019.
[46] D. D. Lee and H. S. Seung, "Learning the parts of objects by nonnegative matrix factorization," *Nature*, vol. 401, no. 6755, pp. 788–791, Oct. 1999.



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